

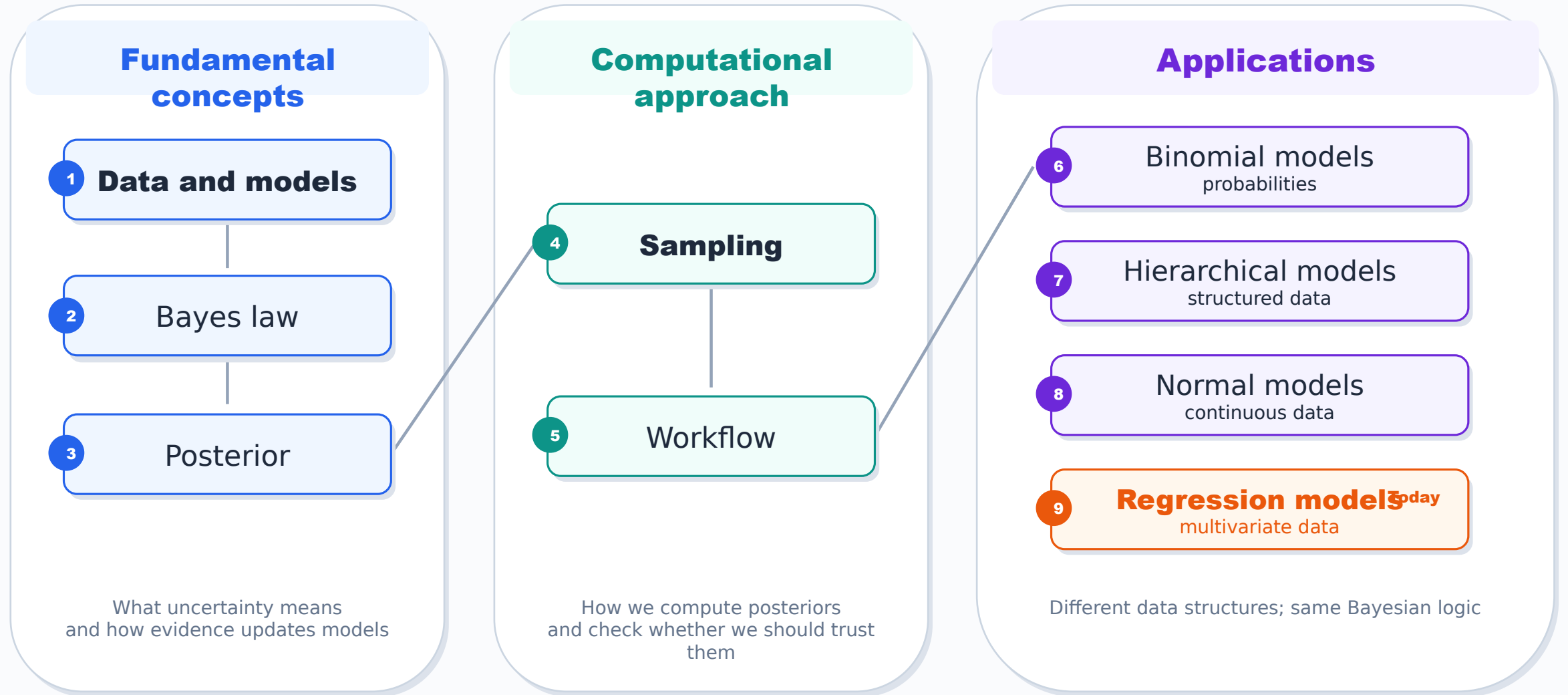
Statistics
367-1-4361
Linear Models
Opher Donchin

8.1 Bivariate data

Examples of how two variables can be related

What we've learned so far

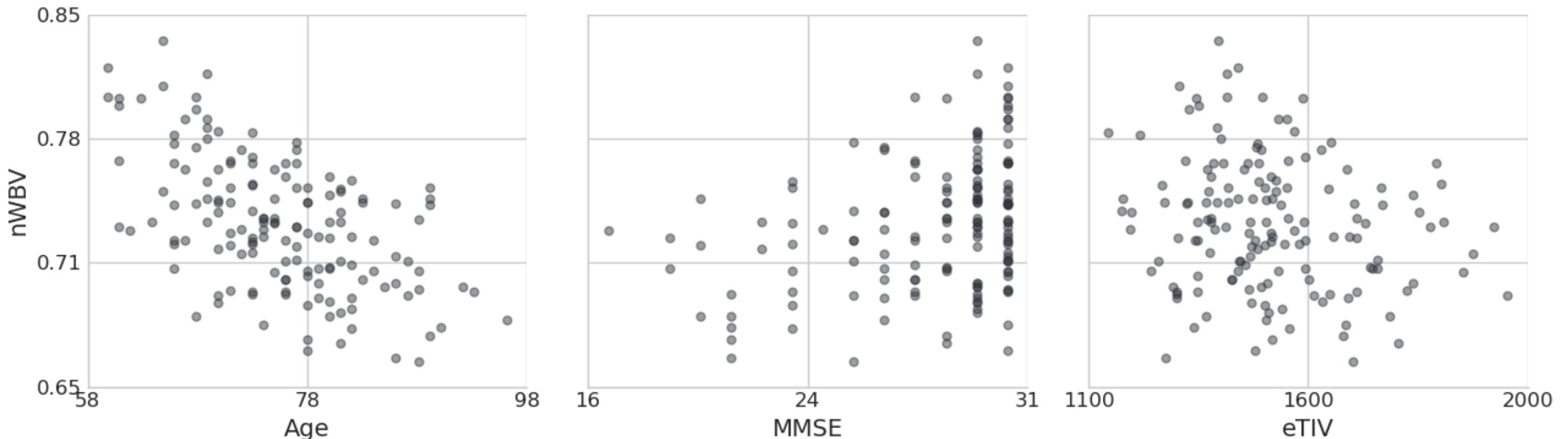
The course has built a chain of ideas, tools, and model families



Today's models relate age, brain volume, and cognitive function

- Characterize
- Estimate parameters
 - Assess effect sizes
- Make inferences

Variable	Meaning	Units
nWBV	Normalized whole-brain volume	Fraction
Age	Age at first MRI visit	years
MMSE	Cognitive screening score; higher is better	points
eTIV	Estimated total intracranial volume	cm ³

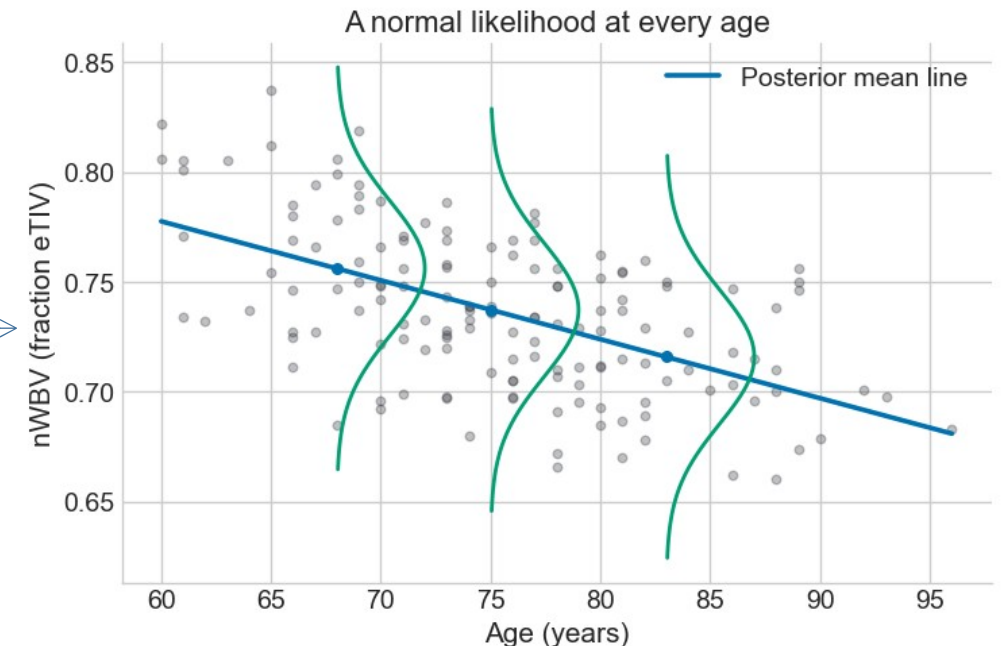
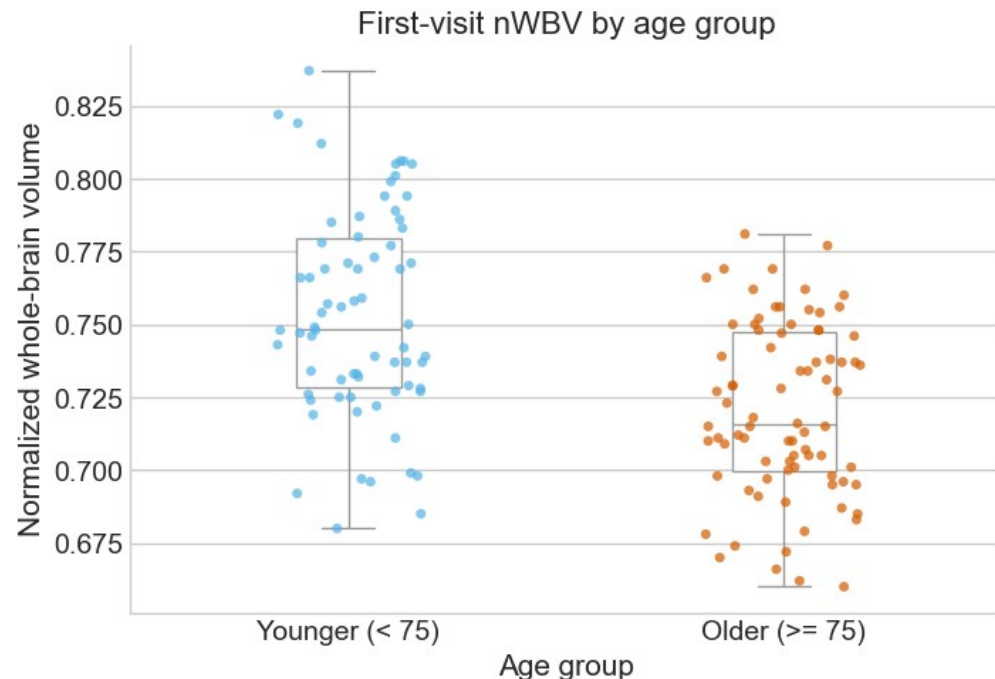


8.2 The bivariate model

Independent variables, dependent variables,
linear model, shared variance

Generalizing the two group normal model

- In two group model
 - Each data point had one of two means
- In bivariate model
 - Each data point has its own mean

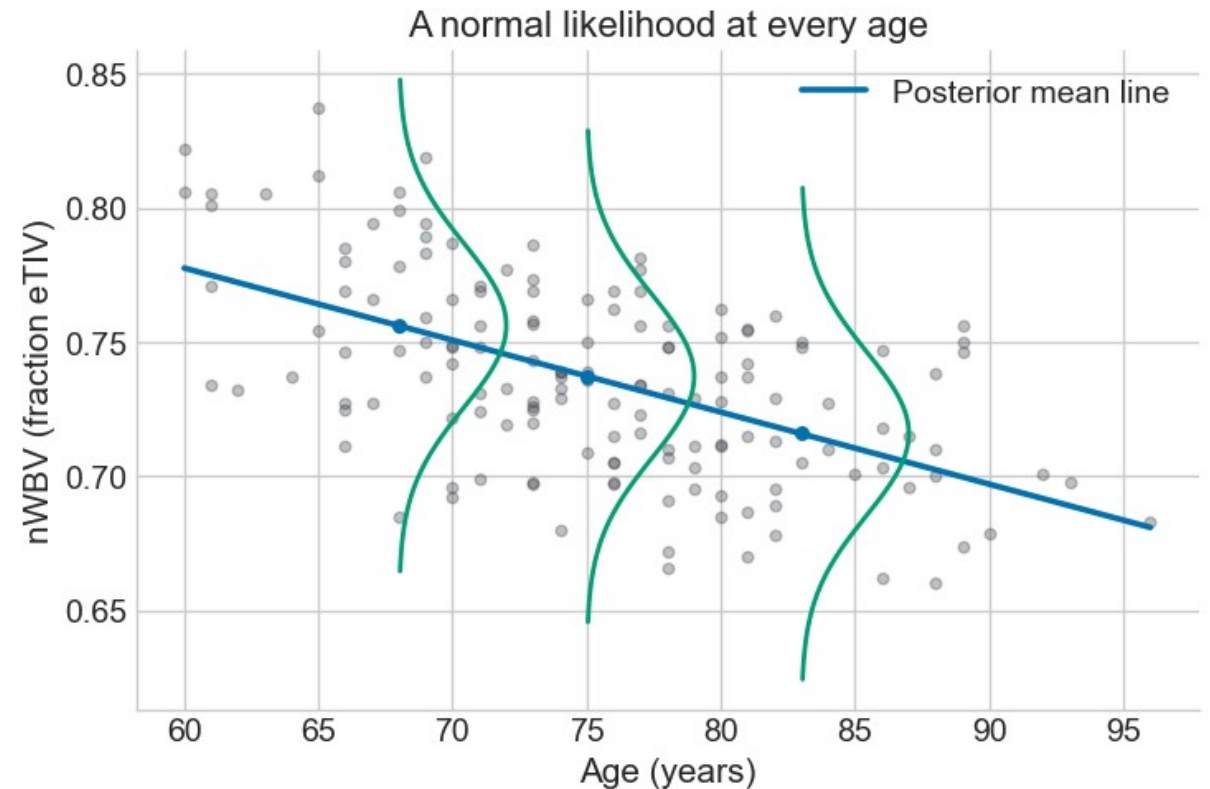


Linear regression has a normal likelihood

- Each data point has its own mean
 - And the equation for the mean is linear
- And they share a standard deviation

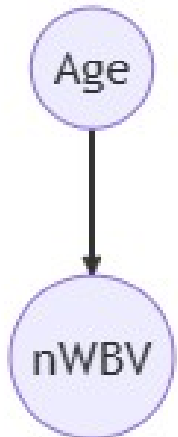
$$\mu_i = \beta_0 + \beta_1 x_i$$

$$y_i \sim N(\mu_i, \sigma)$$



There is an asymmetry in the model

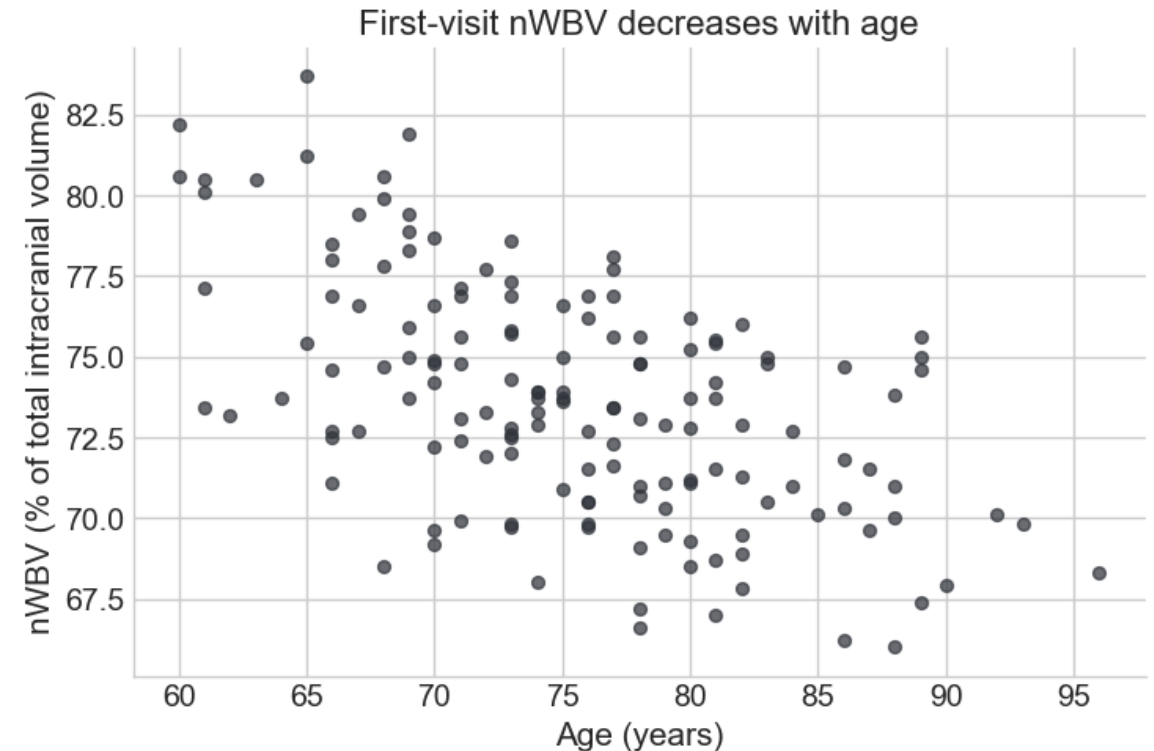
- x : independent variable
 - Assumed to be known precisely
- y : dependent variable
 - Measured with noise
- Implied underlying causal structure



This is called a DAG and they are used to show causal relationships between variables

$$\mu_i = \beta_0 + \beta_1 x_i$$

$$y_i \sim N(\mu_i, \sigma)$$

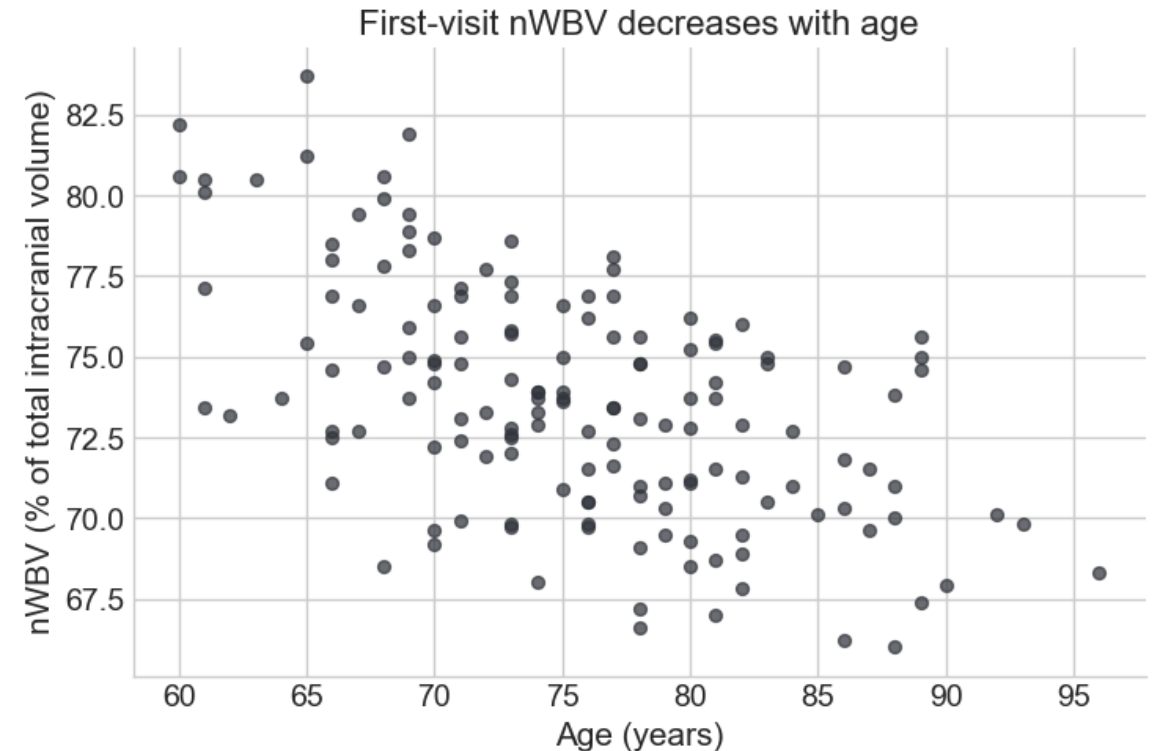


There is an asymmetry in the model

- x : independent variable
 - Assumed to be known precisely
- y : dependent variable
 - Measured with noise
- It can make sense
 - Age affects brain volume
- It can be arbitrary
 - Cognitive test and motor skill test
 - Brain volume and education level

$$\mu_i = \beta_0 + \beta_1 x_i$$

$$y_i \sim N(\mu_i, \sigma)$$



We translate the formal idea into a graphical model

- We have the likelihood
- We need priors

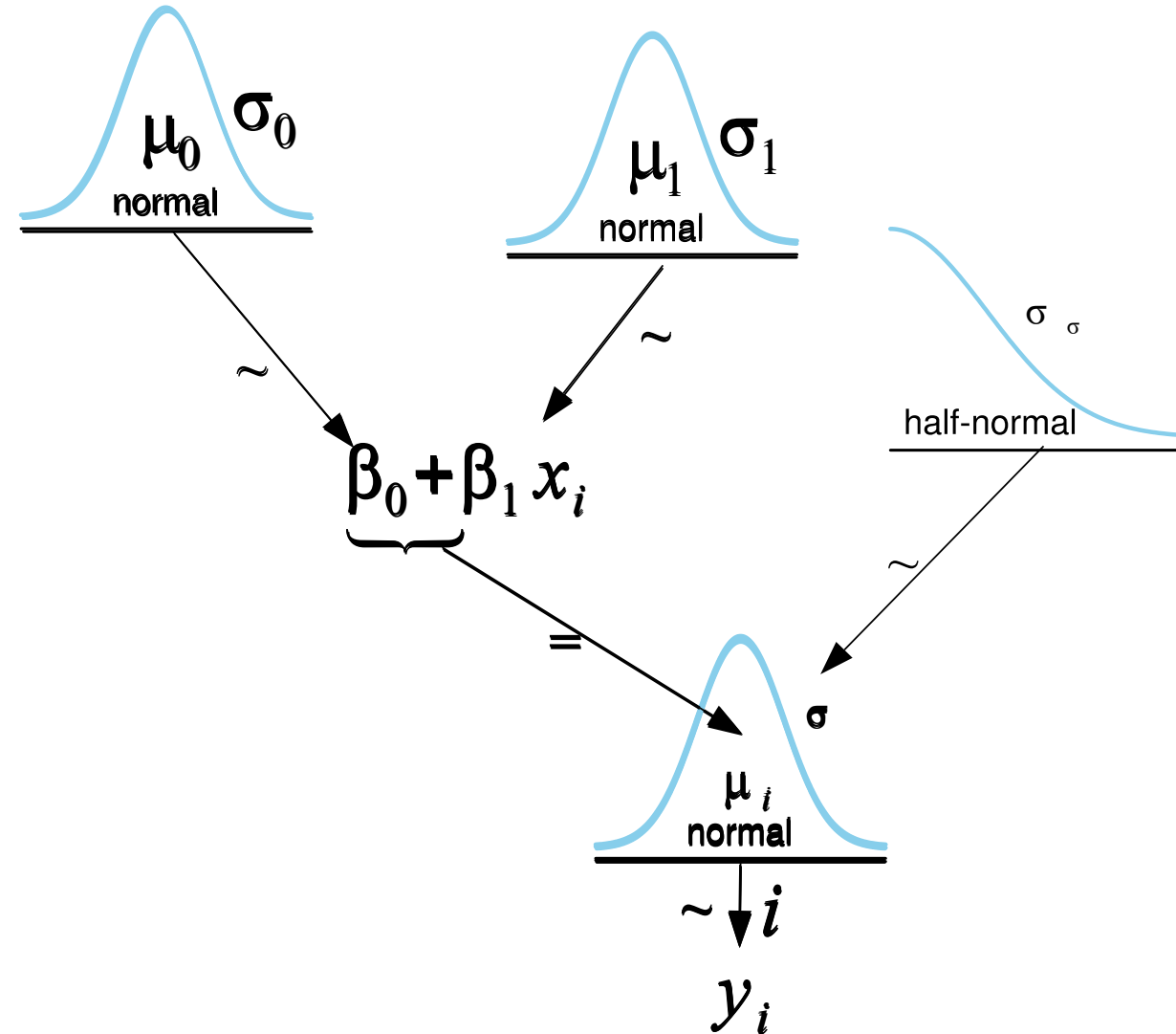
$$\sigma \sim \text{HalfNormal}(\sigma_\sigma)$$

$$\beta_0 \sim N(\mu_0, \sigma_0)$$

$$\beta_1 \sim N(\mu_1, \sigma_1)$$

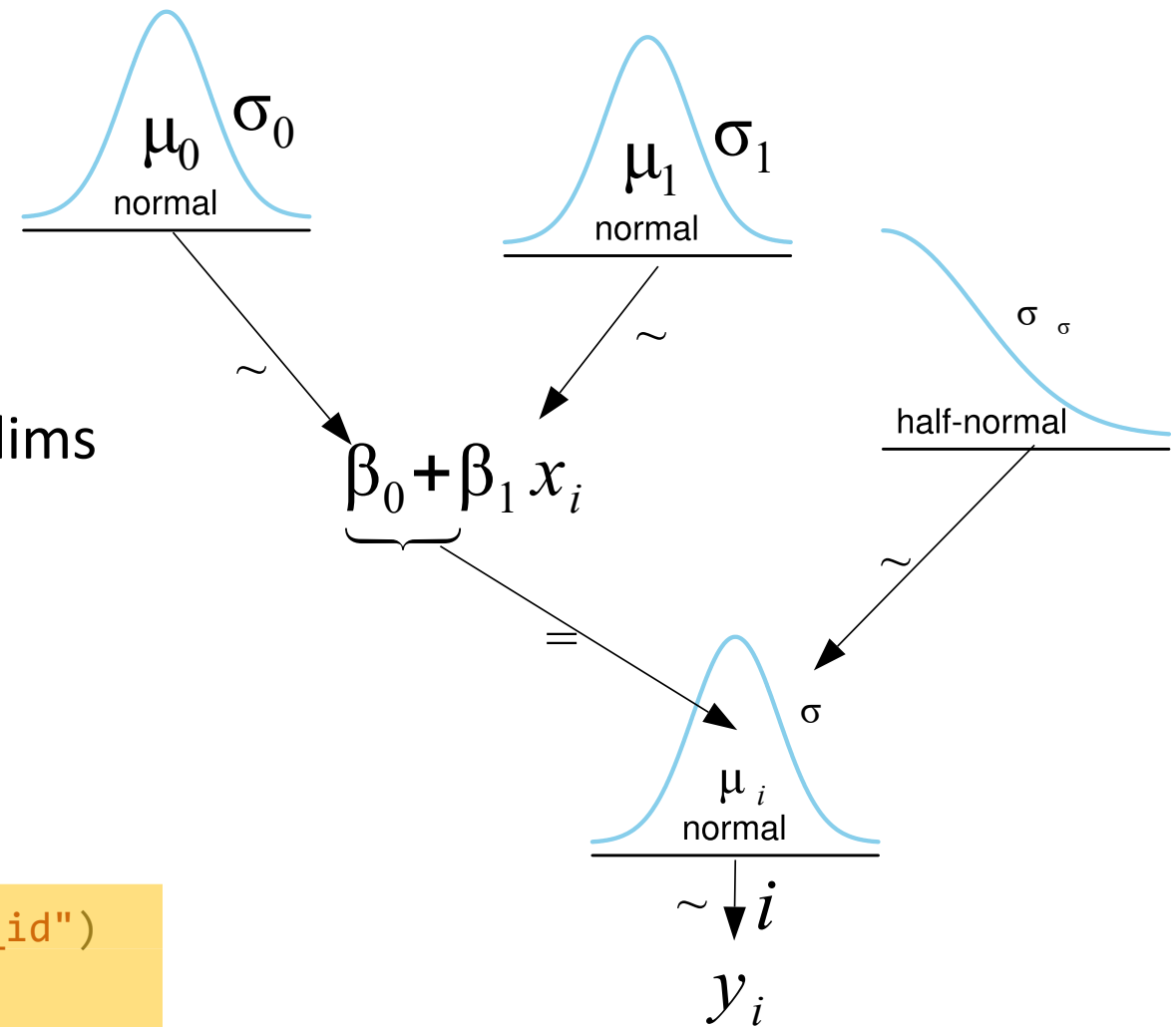
~~$$y_i \sim N(\mu_i, \sigma)$$~~

$$y_i \sim N(\mu_i, \sigma)$$



From graphic model to PyMC

- Automatic broadcasting:
 - Age, mu and nwbv all have the same dims



```
with pm.Model(coords=coords_age) as nwbv_age_model:
```

```
Age_data = pm.Data("Age_data", age_data, dims="obs_id")
```

```
beta0 = pm.Normal("beta0", mu=mu_0, sigma=sigma_0)
```

```
beta1 = pm.Normal("beta1", mu=mu_1, sigma=sigma_1)
```

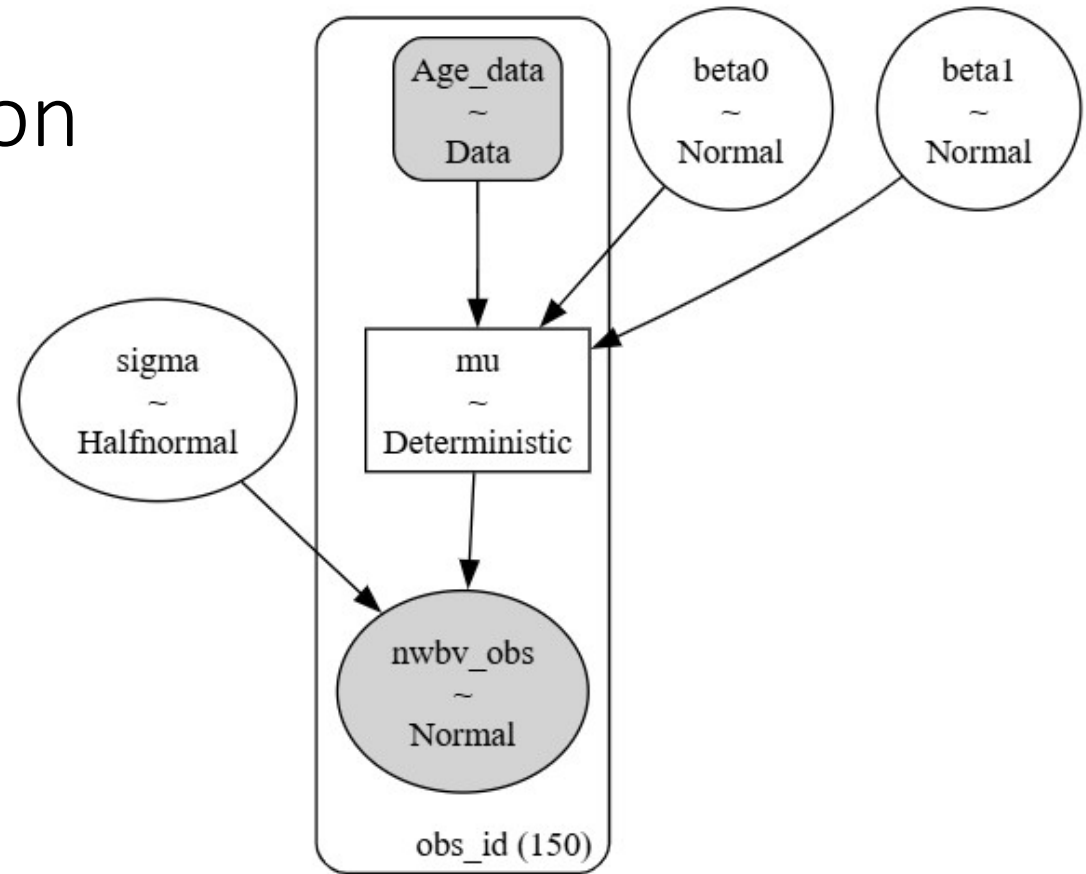
```
sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
```

```
mu = pm.Deterministic("mu", beta0 + beta1 * Age_data, dims="obs_id")
```

```
nwbv_obs = pm.Normal("nwbv_obs", mu=mu, sigma=sigma, observed=nwbv_data, dims="obs_id")
```

From PyMC to graphic representation

- Sanity check for the model

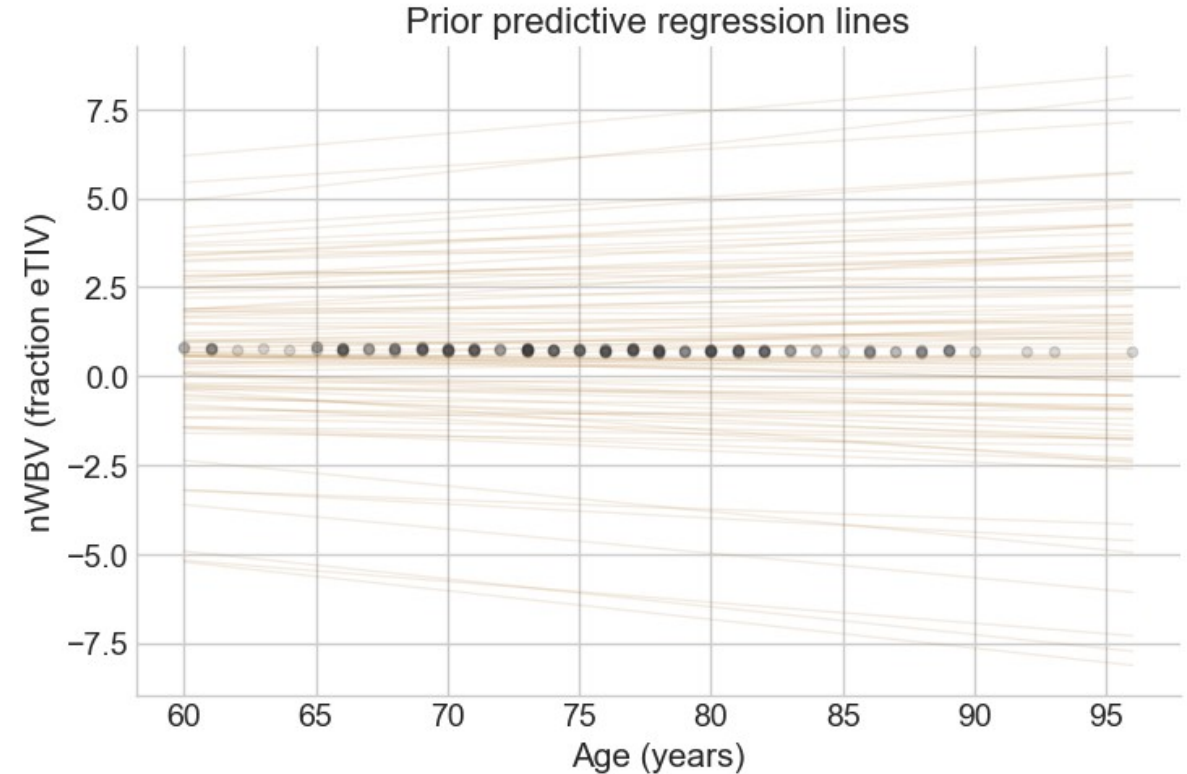


```
with pm.Model(coords=coords_age) as nwbv_age_model:
    Age_data = pm.Data("Age_data", age_data, dims="obs_id")
    beta0 = pm.Normal("beta0", mu=mu_0, sigma=sigma_0)
    beta1 = pm.Normal("beta1", mu=mu_1, sigma=sigma_1)
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu = pm.Deterministic("mu", beta0 + beta1 * Age_data, dims="obs_id")
    nwbv_obs = pm.Normal("nwbv_obs", mu=mu, sigma=sigma, observed=nwbv_data, dims="obs_id")
```

Priors for β_0 and β_1 depend on regression equation

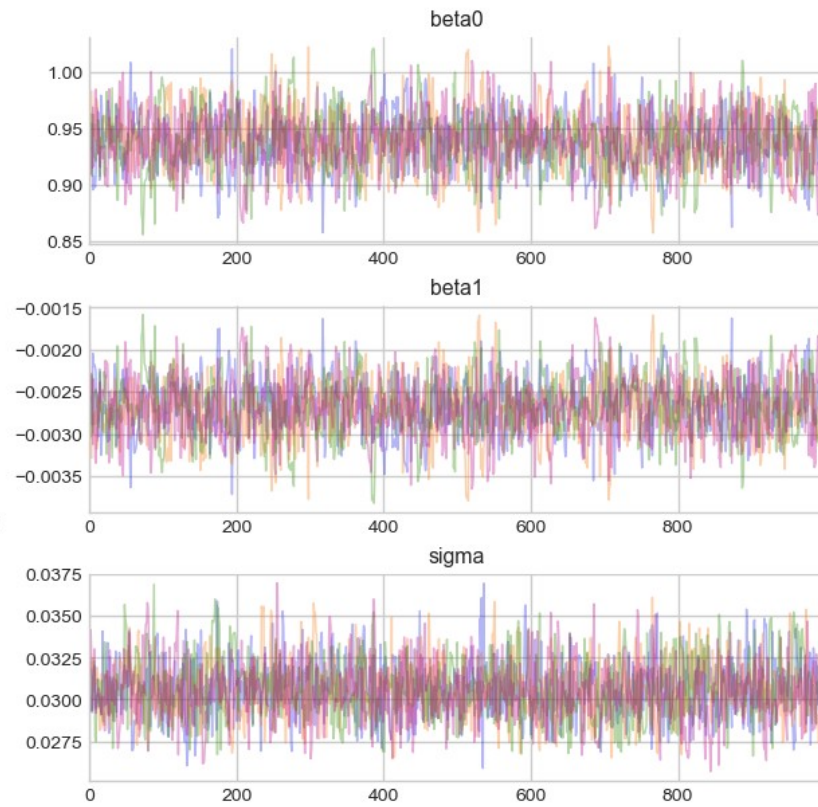
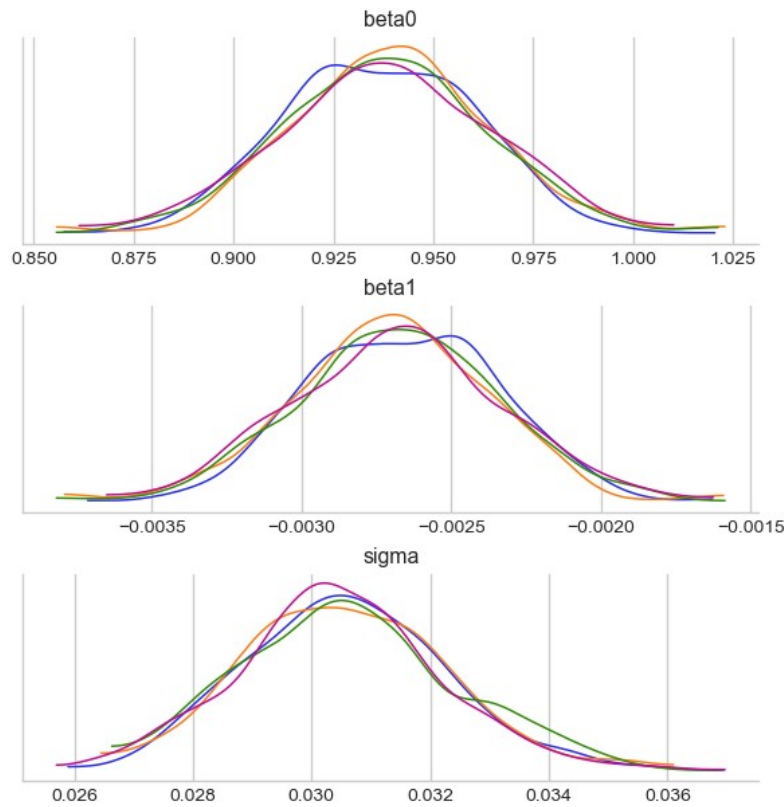
○ Priors for β_0 , β_1 , and σ

- β_1 is the slope: $\frac{\Delta nWBV}{\Delta Age}$
 - nWBV ranges from 0.60 to 0.90
 - Age ranges from 60 to 90
 - Slopes might range from -0.03 to 0.03
 - If we “know” the brain degenerates with age
 - We could have a tighter prior
- β_0 is the intercept: $\mu_{nWBV} - \beta_1 \mu_{Age}$
 - Prior uncertainty in β_0 is influenced by
 - Distance to y axis
 - Uncertainty in β_1



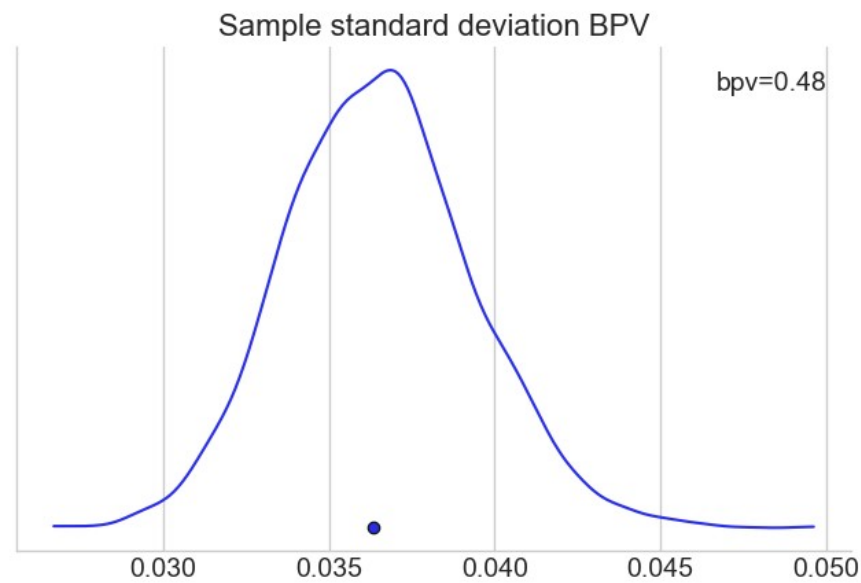
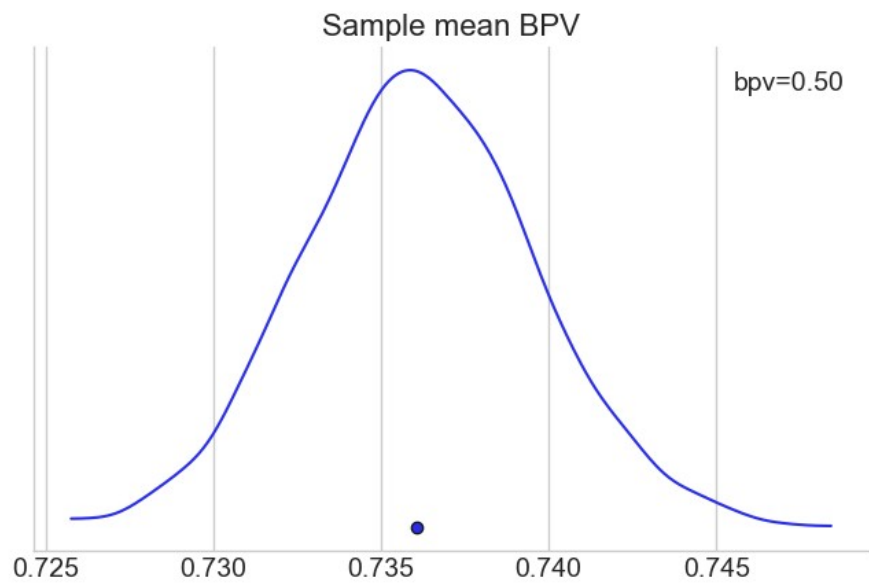
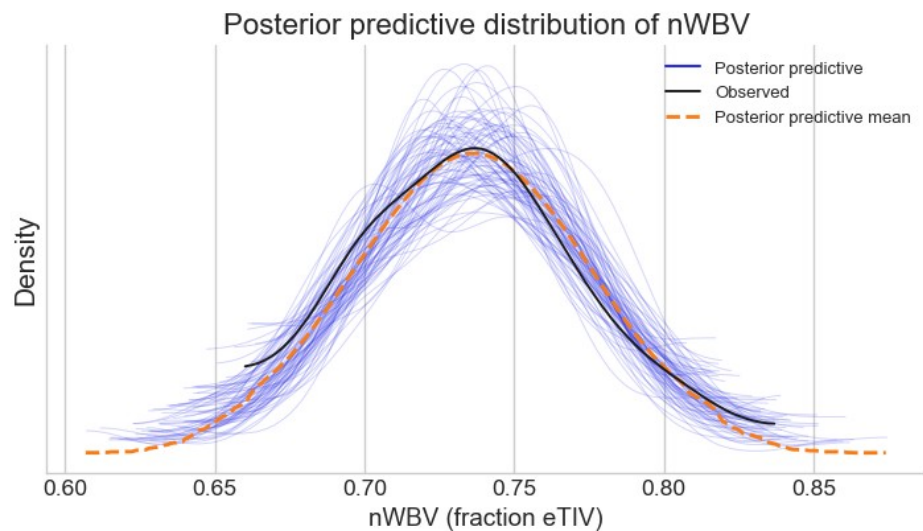
I will proceed with this very broad prior

Sampling looks ok



	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
beta0	0.938	0.026	0.892	0.988	0.001	0.001	1296.0	1485.0	1.0
beta1	-0.003	0.000	-0.003	-0.002	0.000	0.000	1279.0	1513.0	1.0
sigma	0.031	0.002	0.027	0.034	0.000	0.000	1737.0	1581.0	1.0

Posterior predictive distribution looks ok



sample_posterior_predictive can predict on new data

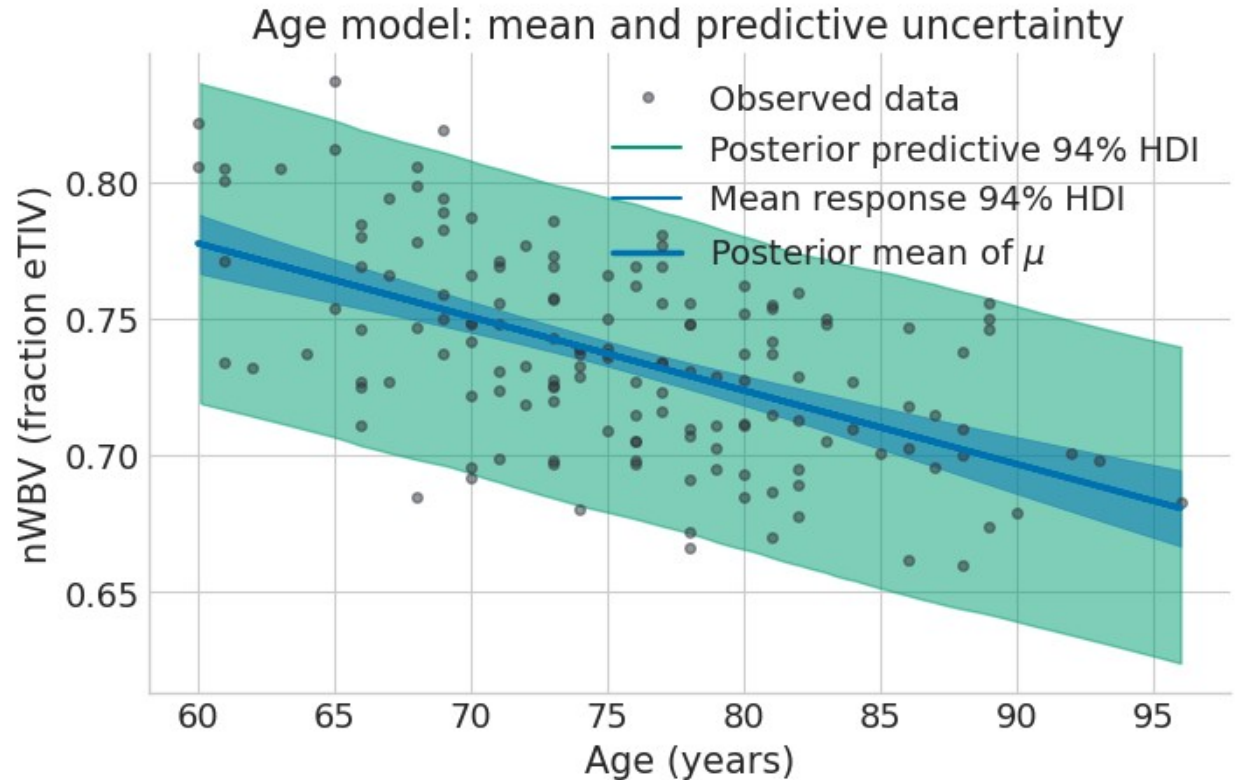
```
age_prediction_idata = sample_model_predictions(  
    nwbv_age_model,  
    nwbv_age_idata,  
    data_updates={"Age_data": age_grid},  
    coords={"obs_id": np.arange(len(age_grid))},  
    restore_data={"Age_data": age_data},  
    restore_coords={"obs_id": np.arange(len(simple_df))},  
)
```

- Need to update coords to appropriate length
- Should restore original data for sanity

```
def sample_model_predictions(model, idata, data_updates, coords, restore_data, restore_coords, var_names):  
    with model:  
        pm.set_data(data_updates, coords=coords)  
        predictions = pm.sample_posterior_predictive(  
            idata,  
            var_names=list(var_names),  
            predictions=True,  
            return_inferencedata=True,  
            random_seed=RANDOM_SEED,  
            progressbar=False,  
        )  
        pm.set_data(restore_data, coords=restore_coords)  
    return predictions
```


Posterior predictive shows model fit

- Predictive data
 - Should look like original data
 - Center and width
 - Should respect model
 - Distance from model
 - Across the range

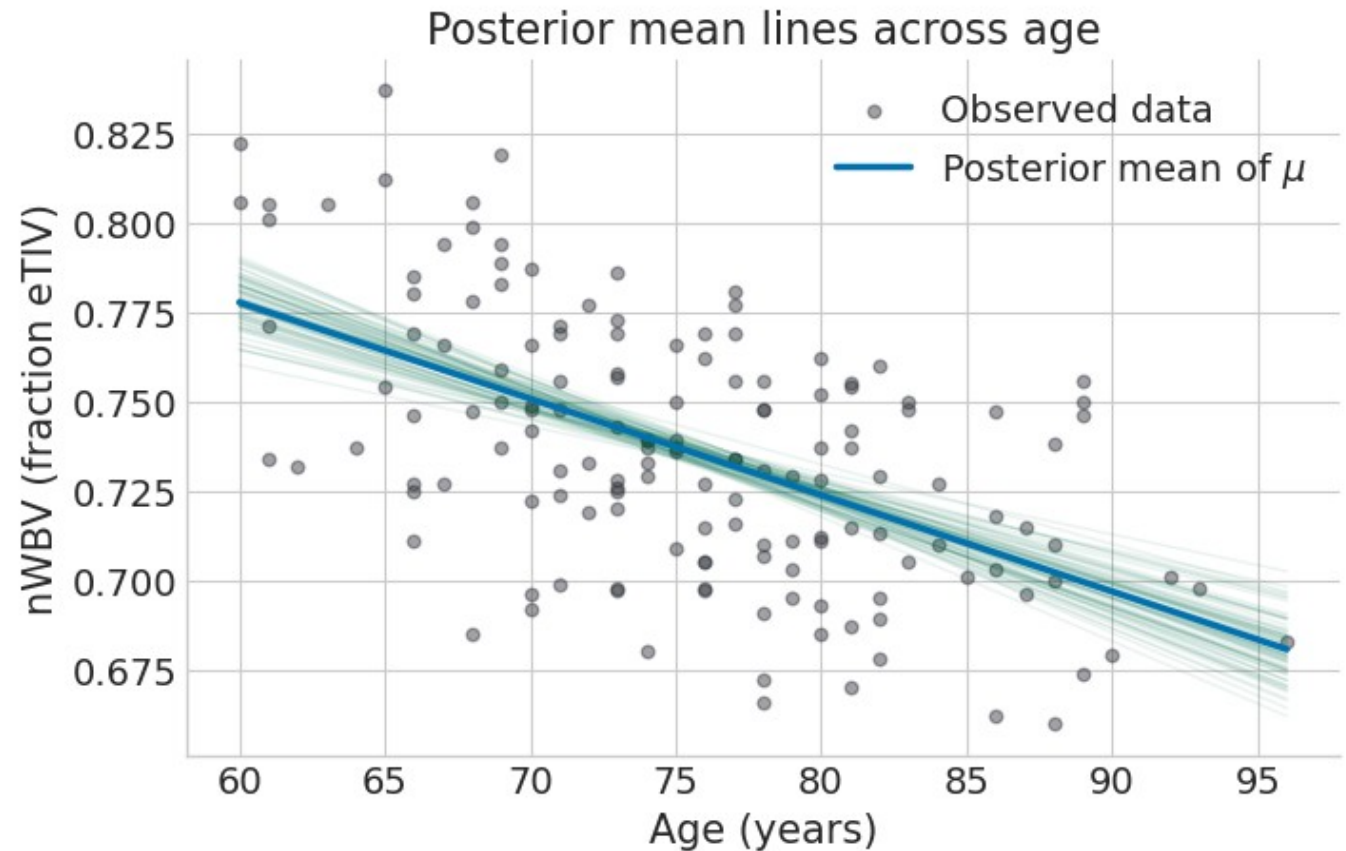


8.3 Posterior regression parameters

Presentation and interpretation

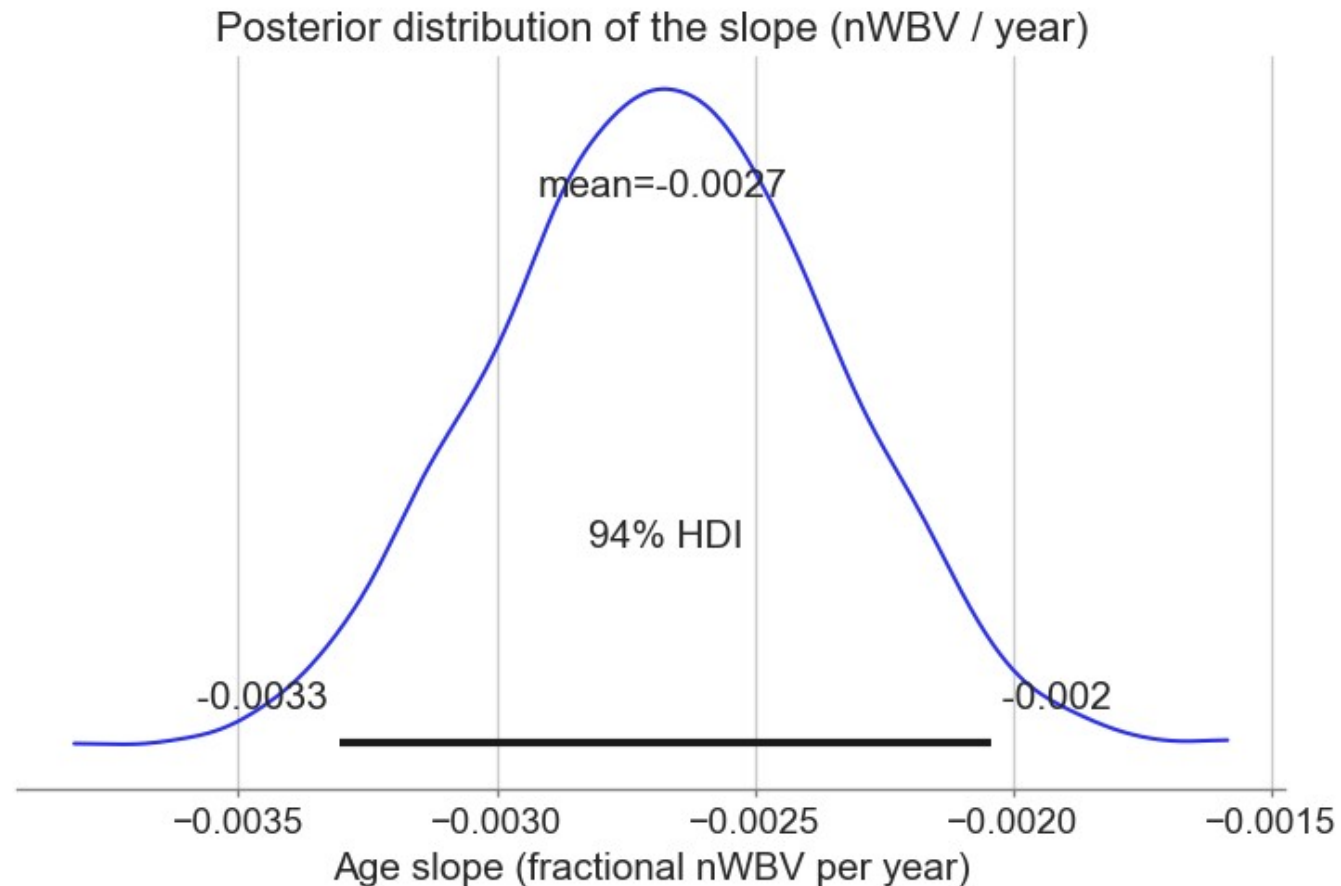
Each posterior sample is a regression line

- Each posterior sample:
 - Has β_0, β_1, σ
 - Defines a possible line
- Near middle of the data
 - More certainty
- At edges
 - Less



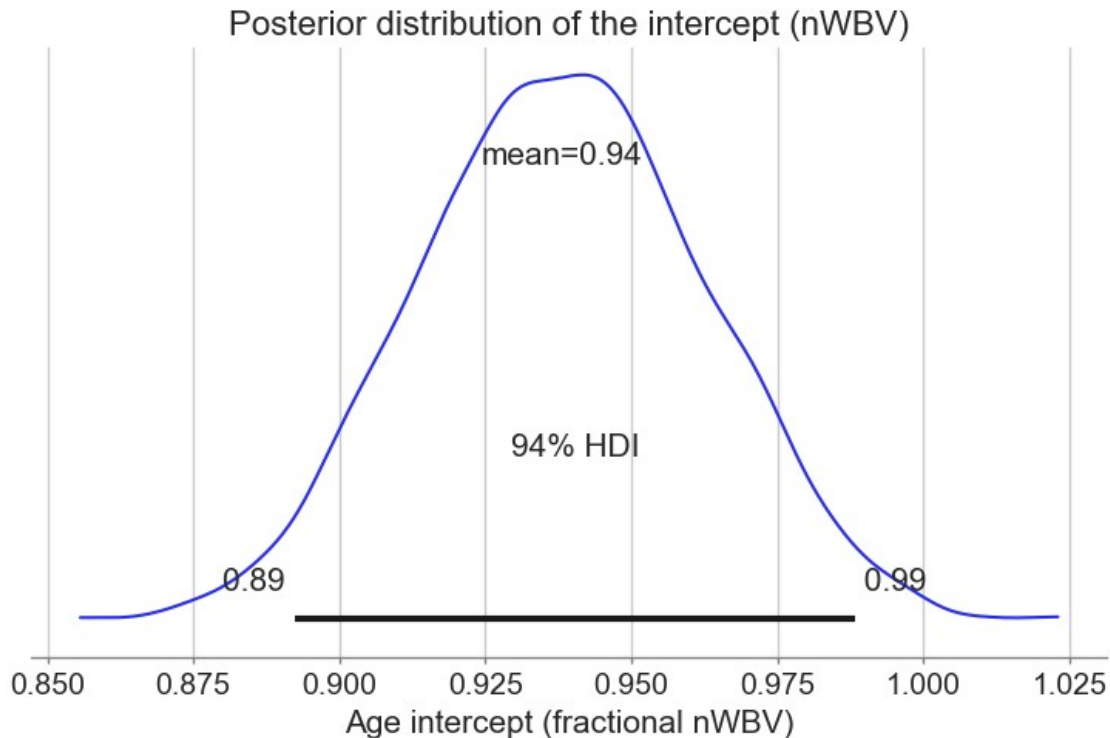
Our main interest is in the slope

- Near 1% loss every 3 years
 - Between 0.20% and 0.33%

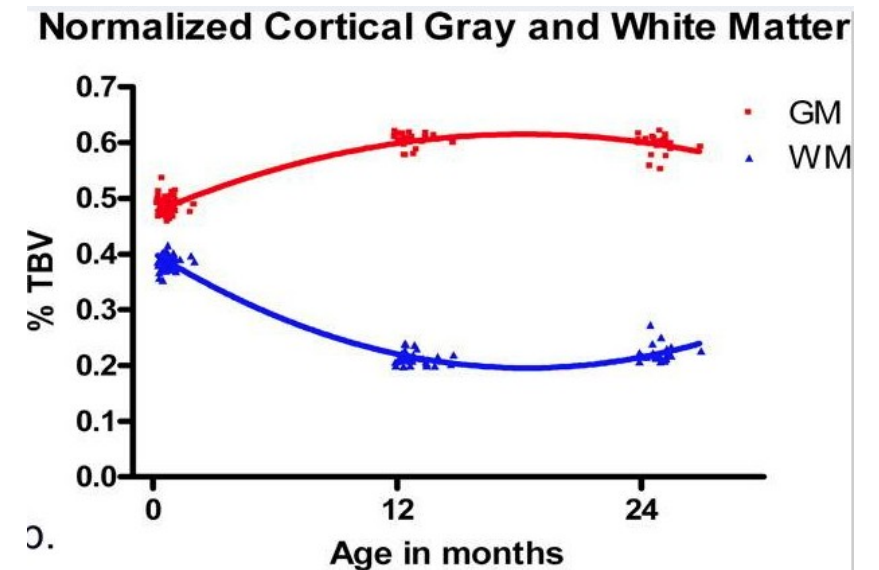


The intercept is nWBV at age 0

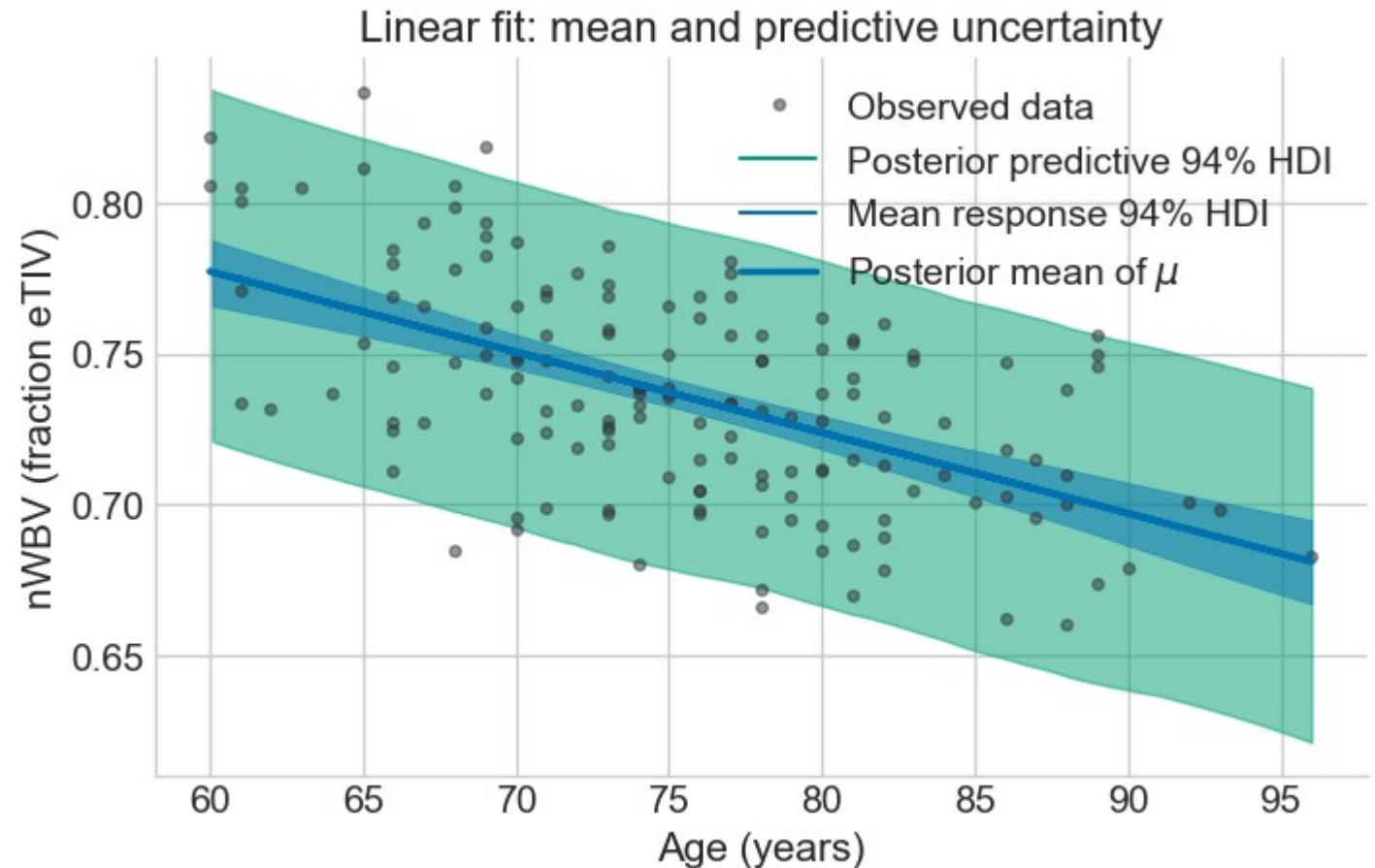
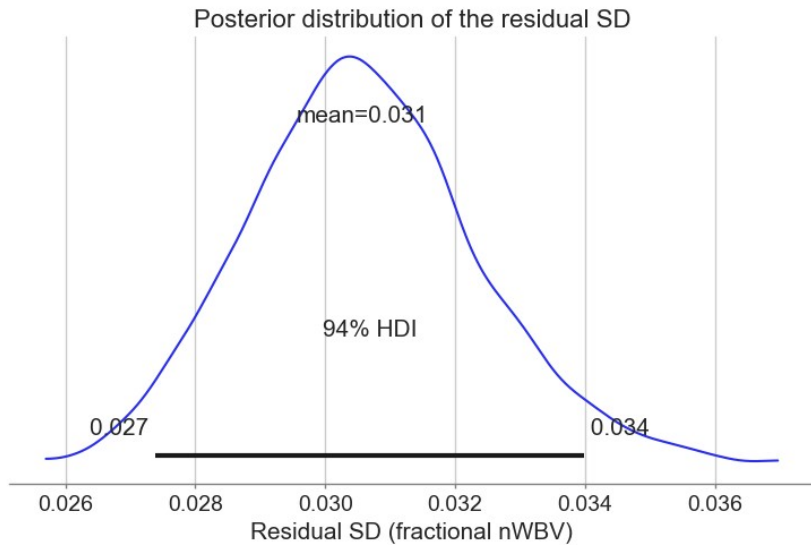
- Model suggests brain volume at birth is 89-99% of cranial volume
 - Broad range
 - Derived from the trend between ages 60 and 90
 - Not necessarily relevant to other ages
 - Regression does not always support extrapolation



Actual nWBV for newborn infants is above 90%

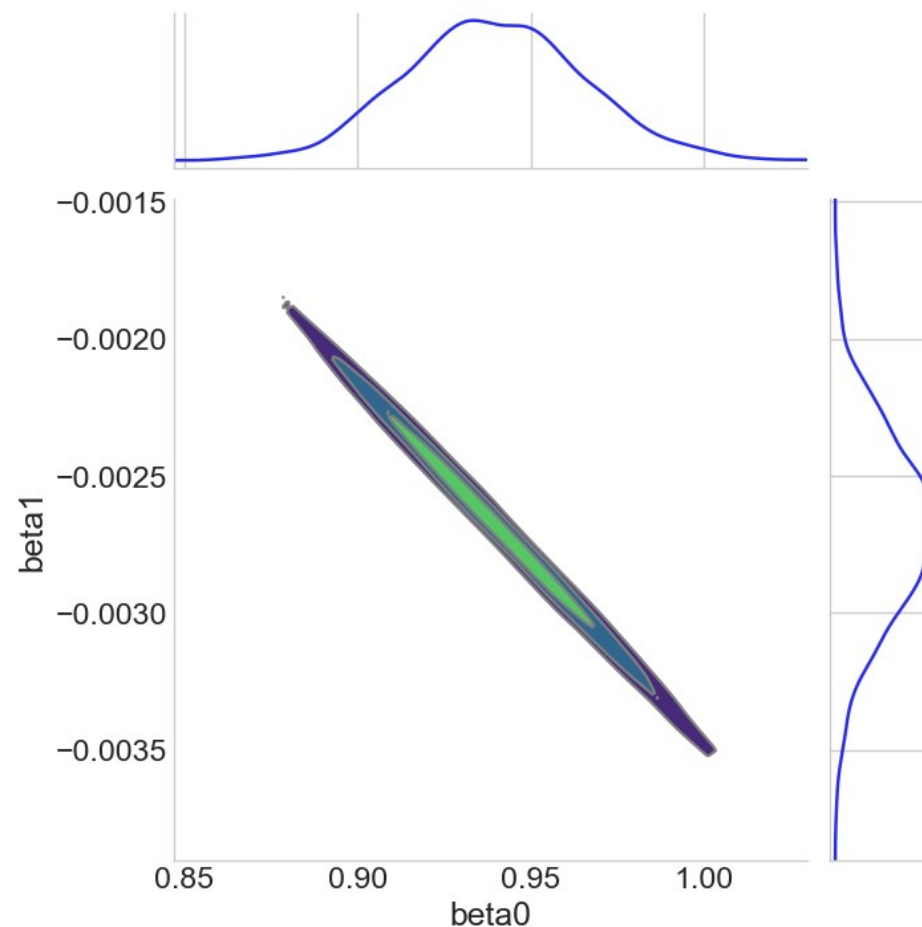


Sigma is the variability among individuals



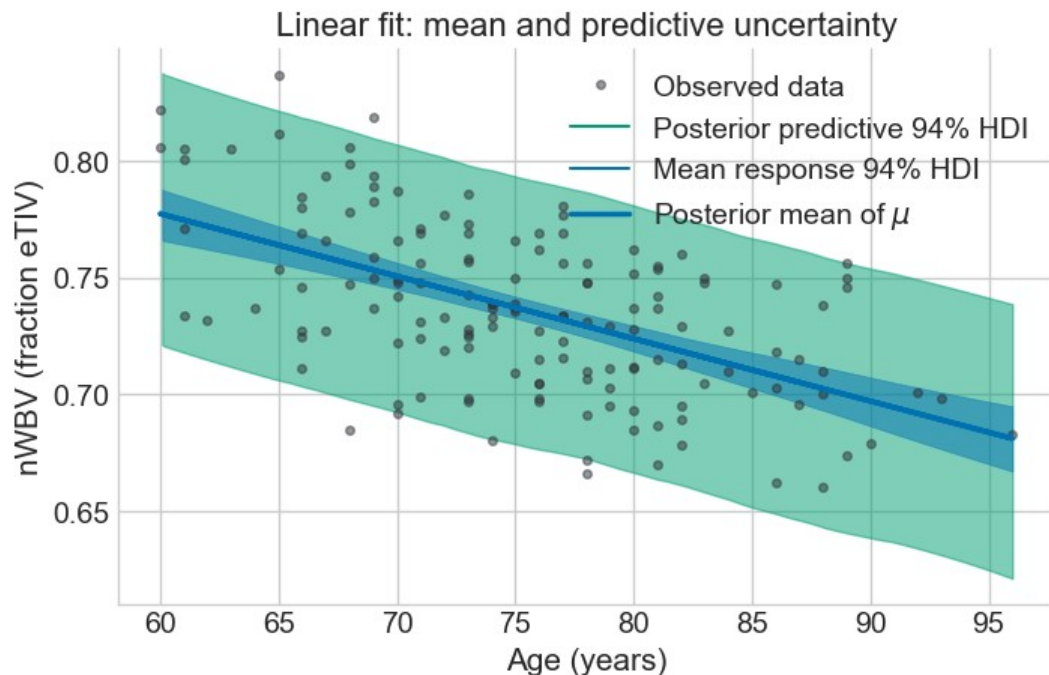
β_0 and β_1 are strongly inversely correlated

- Large negative slope
 - High intercept
- Small negative slope
 - Low intercept

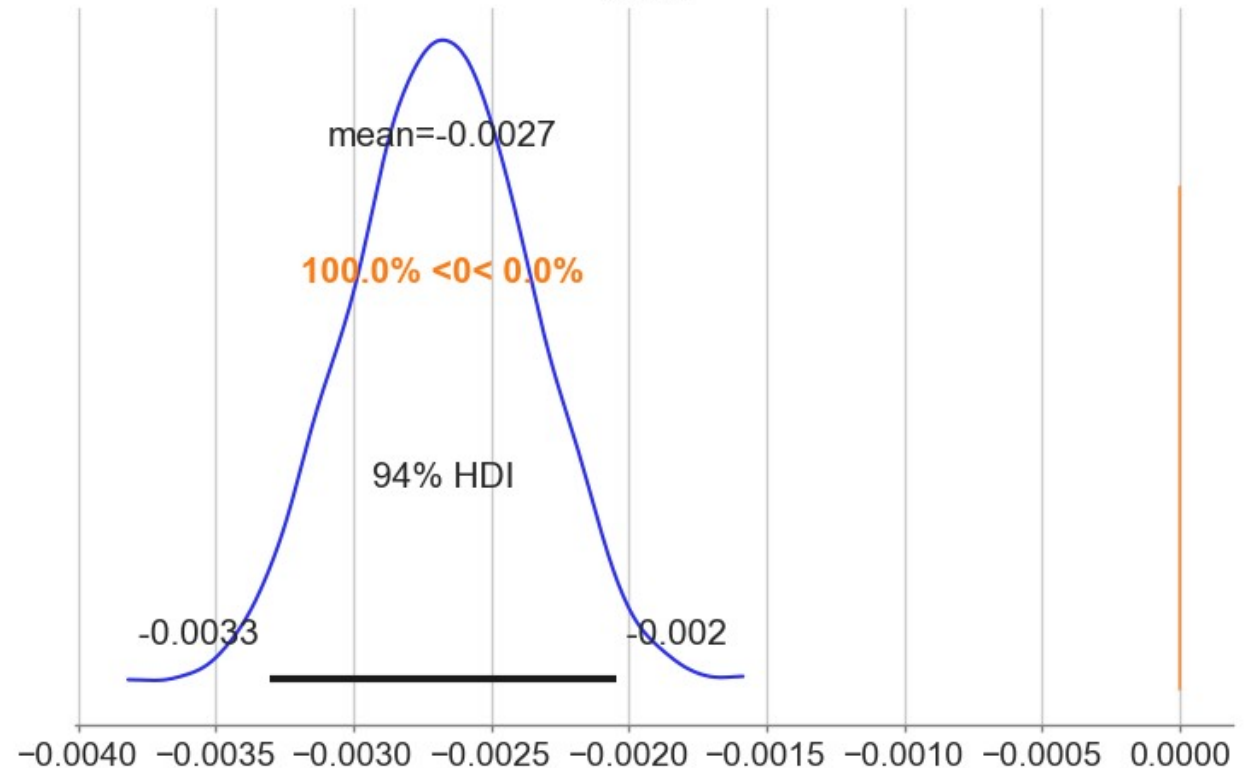
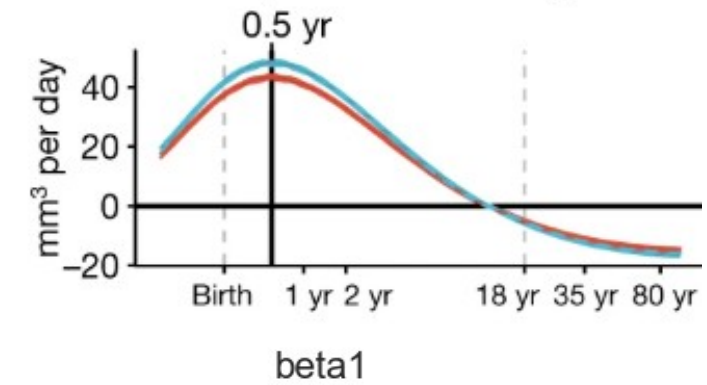


Raw effect size is the slope

- 0.27% from a baseline of 75%
 - 0.36% brain loss per year



Rate of change of brain volume throughout life

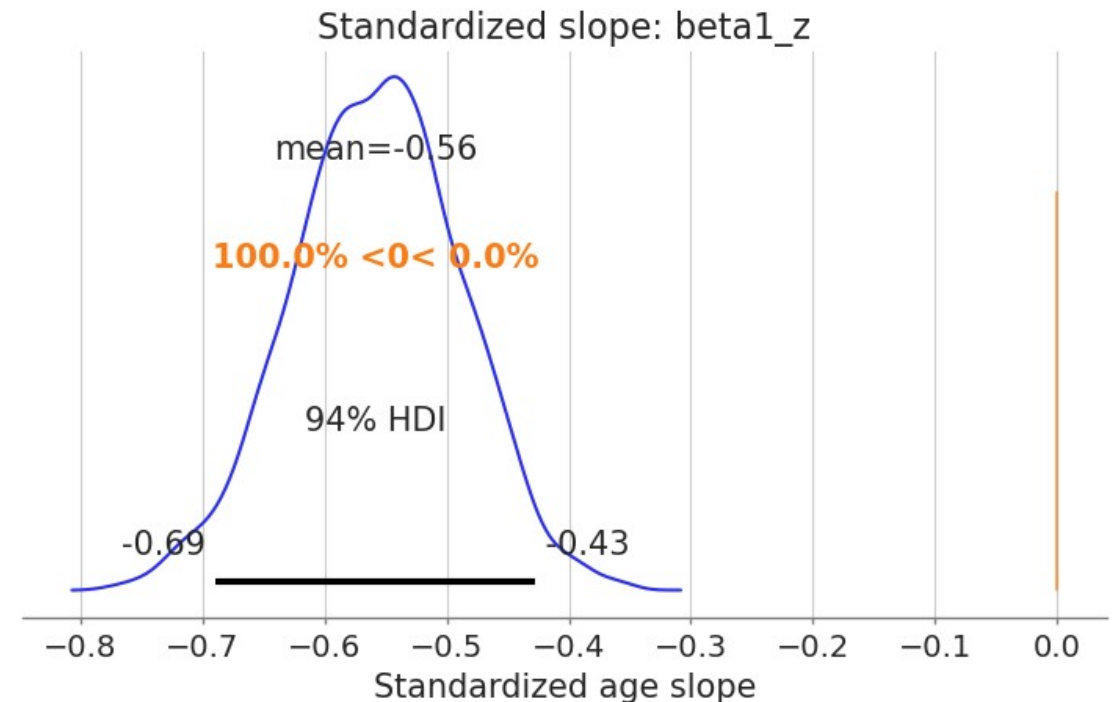


Standardized slope is a standardized effect size

- Standardization
 - Dimensionless units of std
- $\frac{\Delta y}{\Delta x}$ but in std units
- Equal to r
 - The correlation coefficient

$$\beta_1^z = \frac{s_x}{s_y} \beta_1 = r$$

Effect size	Effect size	$\beta_{1, std}$
0.1	Small	
0.3	Medium	
0.5	Large	



8.4 Centered and uncentered data

Moving the center to 0

Centering and standardization remove means and units

Random variable

Data

The variable: x

$$E[x] = \mu$$

$$\text{var}[x] = \sigma^2$$

$$\frac{1}{N} \sum_{i=1}^N x_i = \bar{x}$$

$$\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 = s^2$$

Centered: x^c

$$x^c = x - \mu$$

$$E[x^c] = 0$$

$$\text{var}[x^c] = \sigma^2$$

$$x_i^c = x_i - \bar{x}$$

$$\frac{1}{N} \sum_{i=1}^N x_i^c = 0$$

$$\frac{1}{N-1} \sum_{i=1}^N (x_i^c - \bar{x})^2 = s^2$$

Standardized: x^z

$$x^z = \frac{x - \mu}{\sigma}$$

$$E[x^z] = 0$$

$$\text{var}[x^z] = 1$$

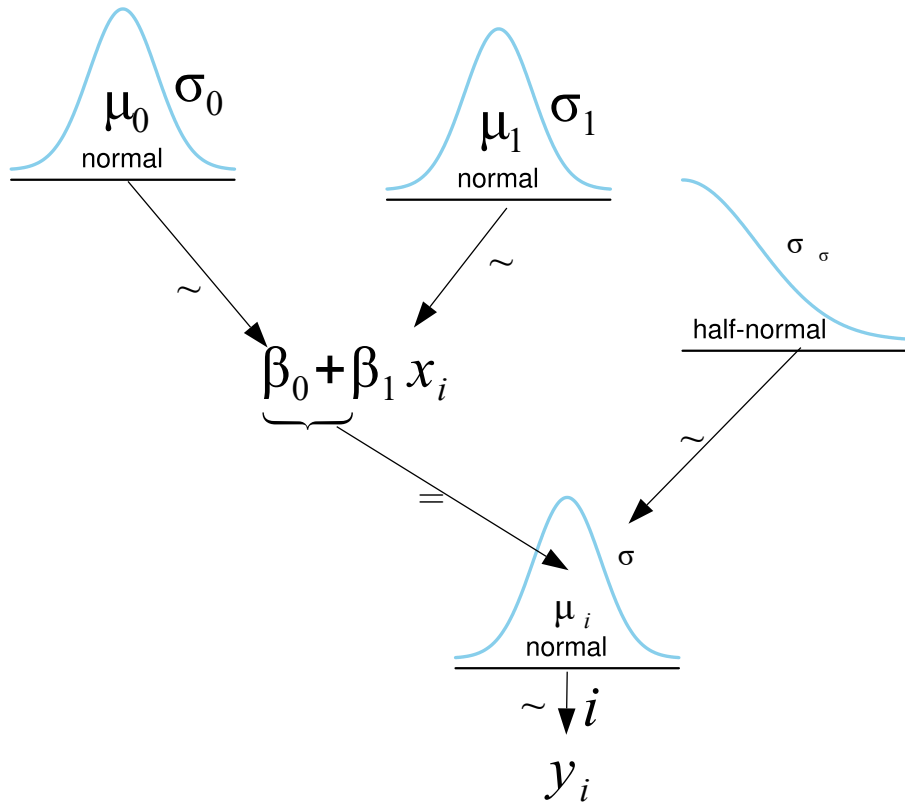
$$x_i^z = \frac{x_i - \bar{x}}{s}$$

$$\frac{1}{N} \sum_{i=1}^N x_i^z = 0$$

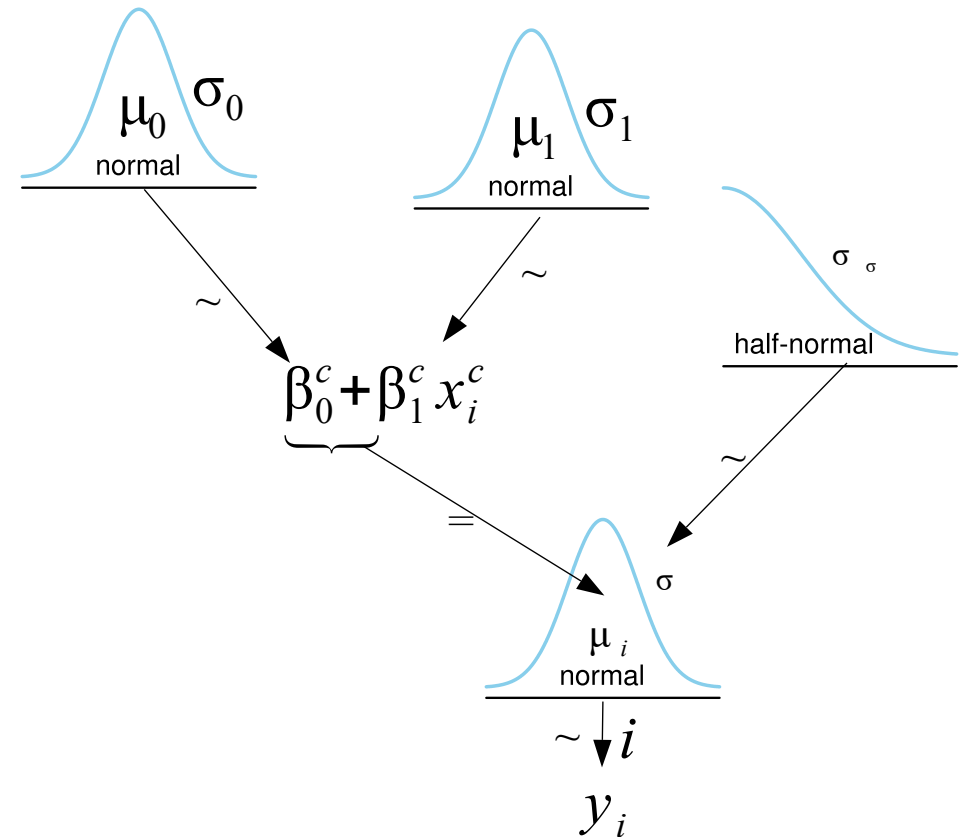
$$\frac{1}{N-1} \sum_{i=1}^N (x_i^z - \bar{x})^2 = 1$$

Switch the model to use a centered independent variable

Uncentered independent variable



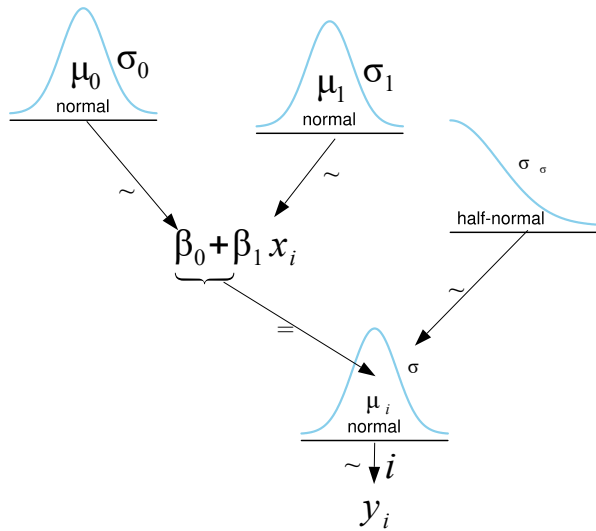
Centered independent variable



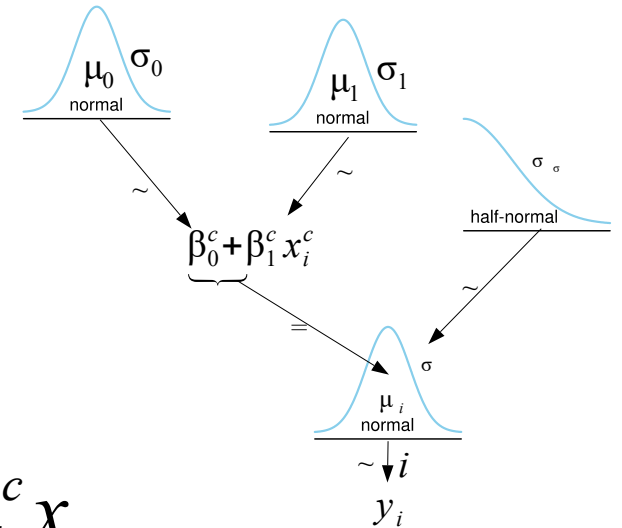
The fit will be the same

- We are just relabeling parameters

Uncentered independent variable



Centered independent variable



$$\mu_i = \beta_0 + \beta_1 x_i = \beta_0^c + \beta_1^c x_i^c$$

$$\beta_0 + \beta_1 x_i = \beta_0^c + \beta_1^c (x_i - \bar{x})$$

$$\beta_0 + \beta_1 x_i = (\beta_0^c - \beta_1^c \bar{x}) + \beta_1^c x_i$$

$$\beta_1 = \beta_1^c$$

$$\beta_0 = \beta_0^c - \beta_1^c \bar{x}$$

Prior for β_0 will be centered at $y(\bar{x})$

- The “intercept” when $x^c = 0$ is equal to μ_i when $x_i = \bar{x}$
 - In most cases, this will be $\bar{y}$

$$\begin{array}{lll} \beta_1 = \beta_1^c & \mu_i = \beta_0^c + \beta_1 x_i^c & 0 = x_i^c = x - \bar{x} \\ \beta_0 = \beta_0^c - \beta_1 \bar{x} & y_i \sim N(\mu_i, \sigma) & x = \bar{x} \end{array}$$

Uncentered independent variable

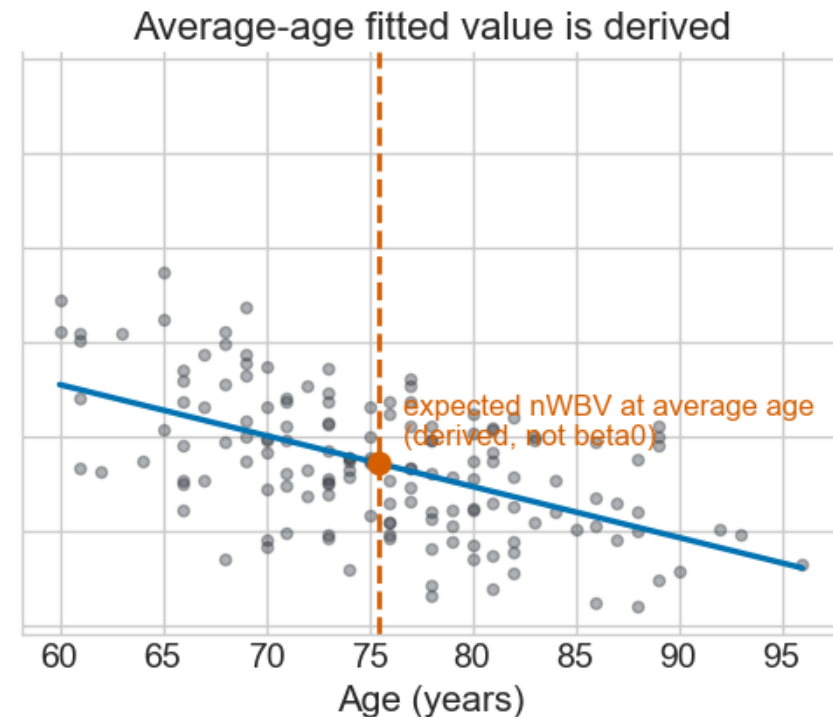
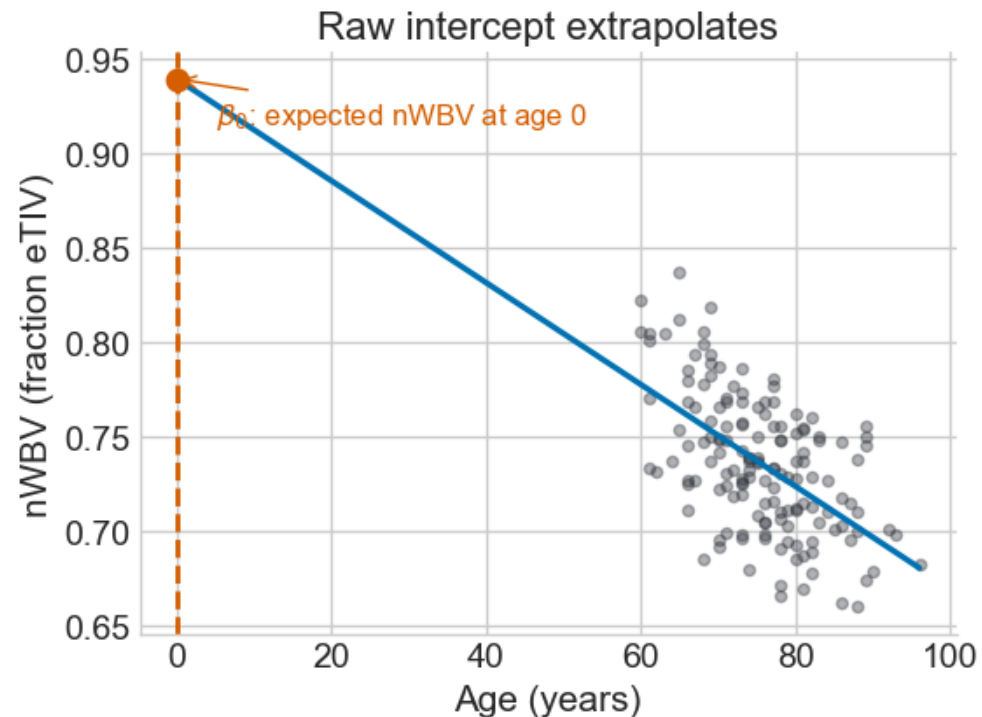
```
mu_0 = np.mean(nwbv_data) - mu_1 * np.mean(age_data)
sigma_0 = 0.5*sigma_1*np.mean(age_data)
```

Centered independent variable

```
mu_0_centered = nwbv_data.mean()
sigma_0_centered = 0.05
```

Intercept has different meaning in the two models

- Scientific question
 - Which do you care about?
- Statistical question
 - Which produces stable inference?
 - Because we can convert between them



The sampling is slower and less independent

Uncentered independent variable

4 chains for 1,000 tune and 1,000 draw iterations

17 seconds

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
beta0	0.939	0.026	0.892	0.987	0.001	0.001	1215.0	1578.0	1.0
beta1	-0.003	0.000	-0.003	-0.002	0.000	0.000	1229.0	1646.0	1.0
sigma	0.031	0.002	0.027	0.034	0.000	0.000	1776.0	1674.0	1.0

Centered independent variable

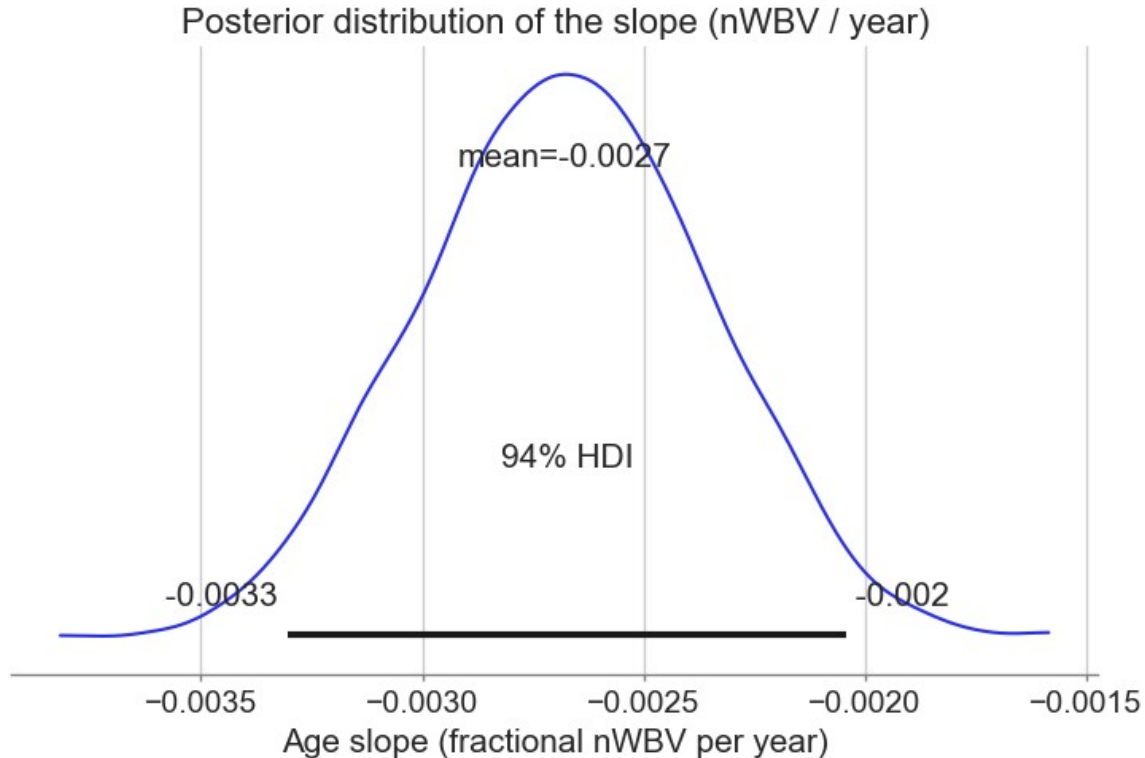
4 chains for 1,000 tune and 1,000 draw iterations

3 seconds

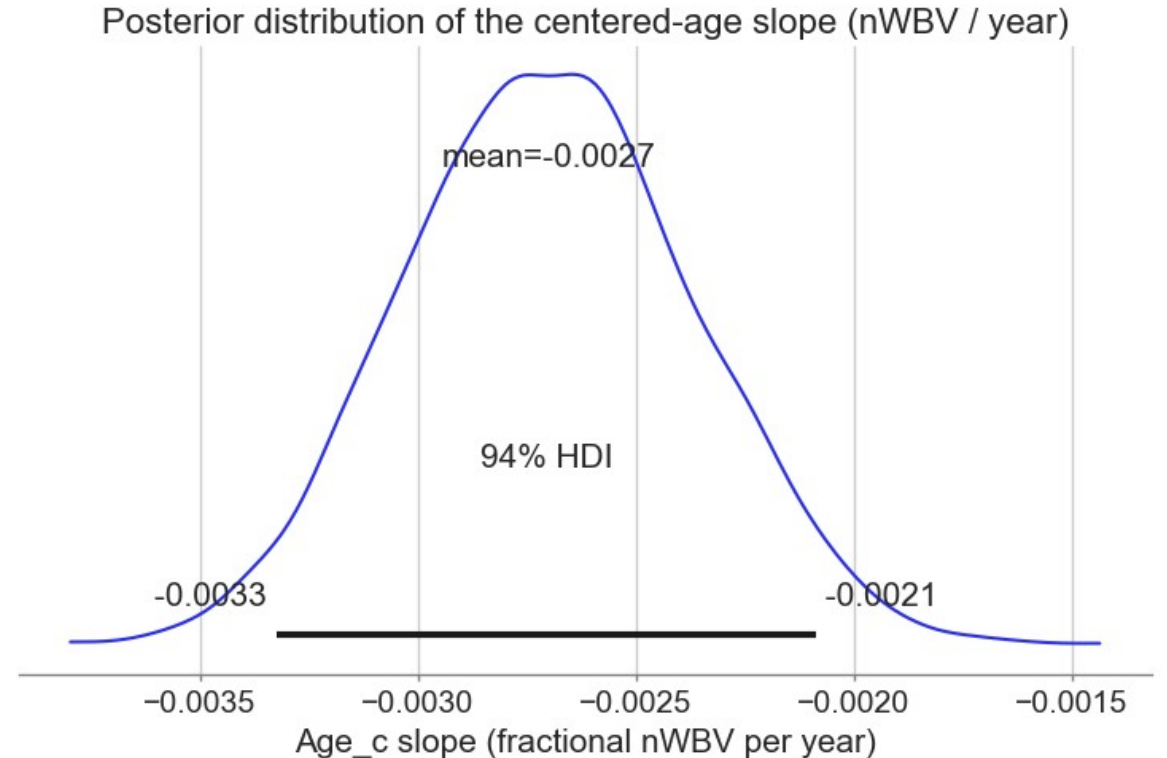
	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
beta0	0.736	0.003	0.731	0.741	0.0	0.0	3920.0	2673.0	1.0
beta1	-0.003	0.000	-0.003	-0.002	0.0	0.0	3759.0	2793.0	1.0
sigma	0.031	0.002	0.027	0.034	0.0	0.0	3430.0	2605.0	1.0

The posterior of the slope is unchanged

Uncentered independent variable

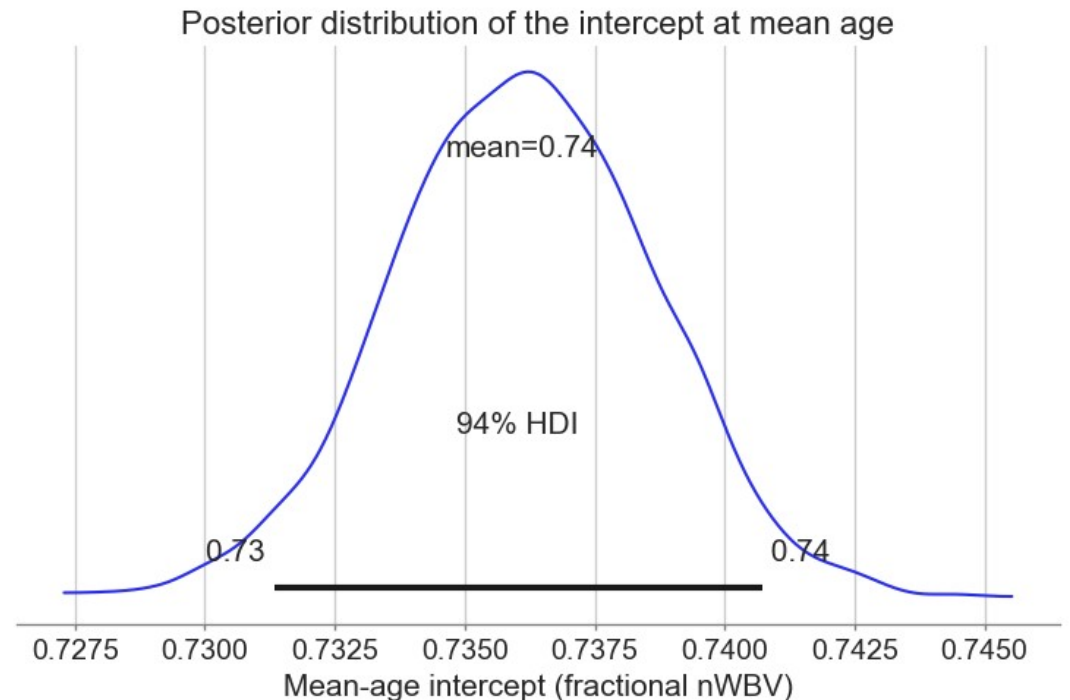
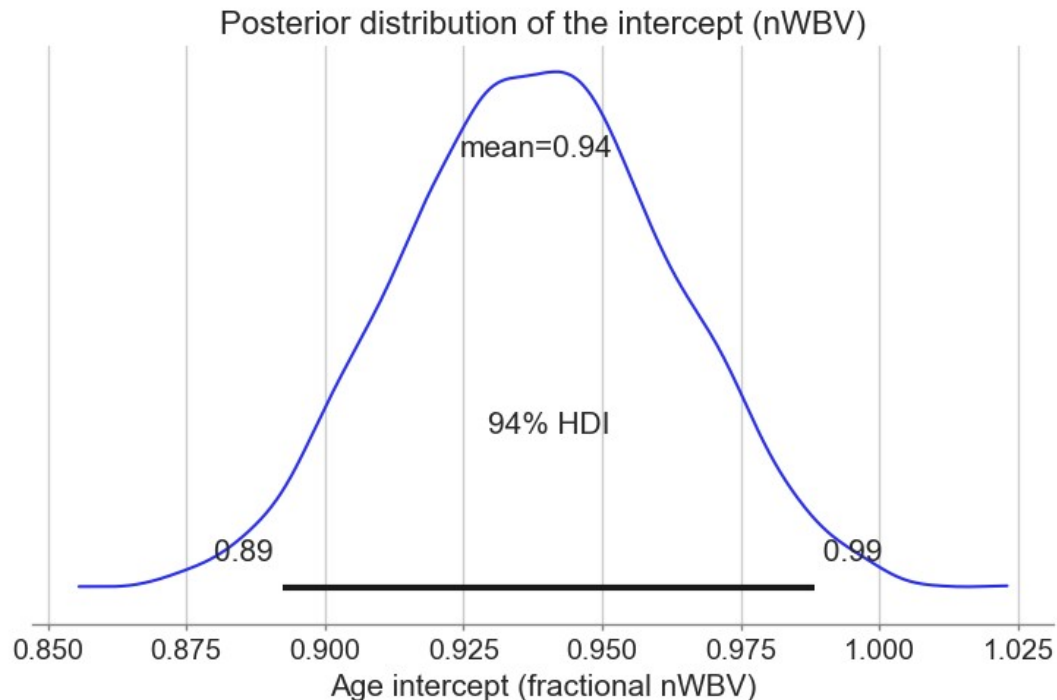


Centered independent variable



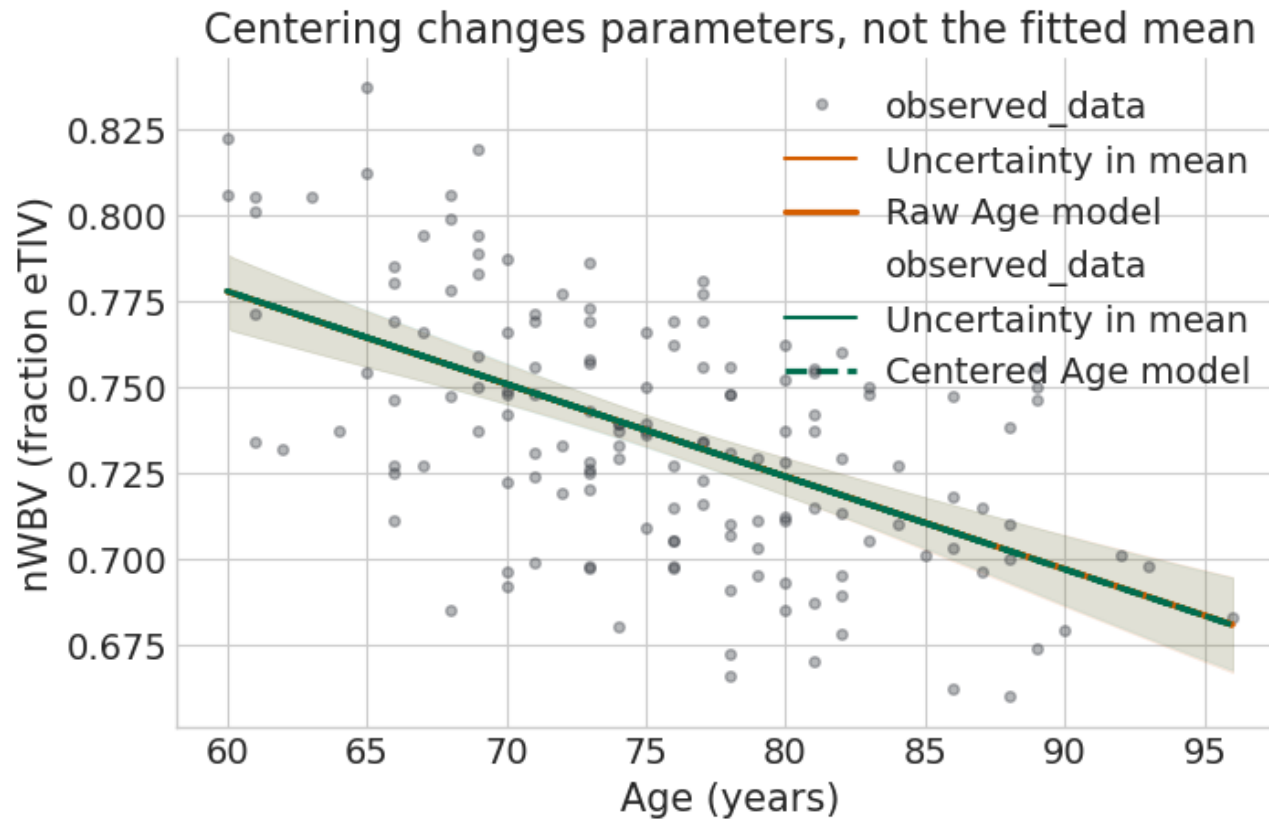
The posterior of the intercept is different

- y value at the mean of x
 - More meaningful and interpretable than extrapolation to 0
 - Often, not always
- Narrower by a factor of 10



The final fit is the same

- Reparameterization affects sampling and interpretability of parameters
 - Not fit



8.5 Multivariate regression

Multivariate dependencies, the multivariate model with simple and matrix variables, workflow

Mini-mental state exam (MMSE)

- Widely used in clinical practice and in research

Mini-Mental State Examination (MMSE)

Patient's Name: _____ Date: _____

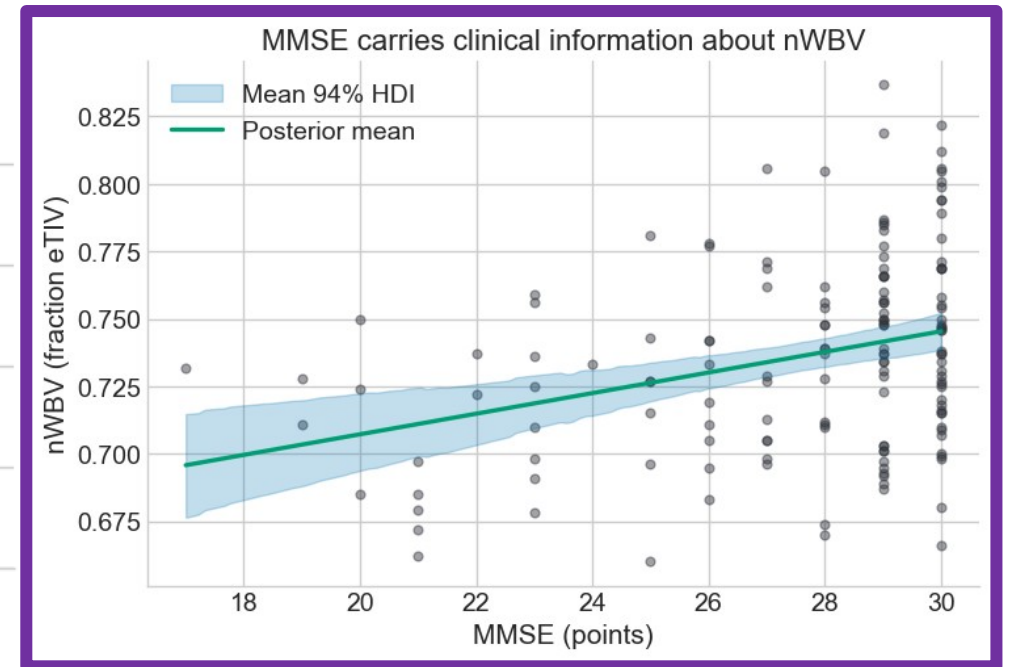
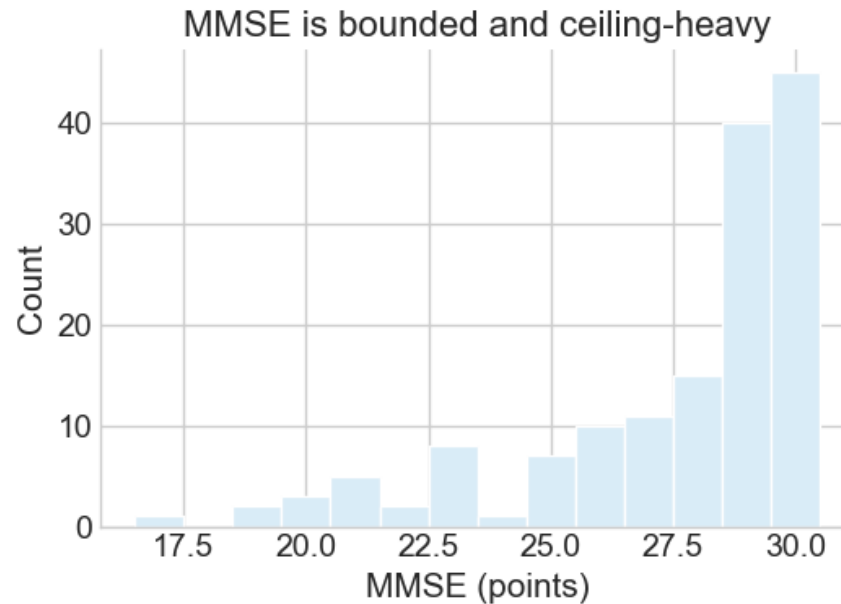
Instructions: Ask the questions in the order listed.
Score one point for each correct response within each question or activity.

Maximum Score	Patient's Score	Questions
5		"What is the year? Season? Date? Day of the week? Month?"
5		"Where are we now: State? County? Town/city? Hospital? Floor?"
3		The examiner names three unrelated objects clearly and slowly, then asks the patient to name all three of them. The patient's response is used for scoring. The examiner repeats them until patient learns all of them, if possible. Number of trials: _____
5		"I would like you to count backward from 100 by sevens." (93, 86, 79, 72, 65, ...) Stop after five answers. Alternative: "Spell WORLD backwards." (D-L-R-O-W)

Method	Score	Interpretation
Single Cutoff	<24	Abnormal
Range	<21 >25	Increased odds of dementia Decreased odds of dementia
Education	21 <23 <24	Abnormal for 8th grade education Abnormal for high school education Abnormal for college education
Severity	24-30 18-23 0-17	No cognitive impairment Mild cognitive impairment Severe cognitive impairment

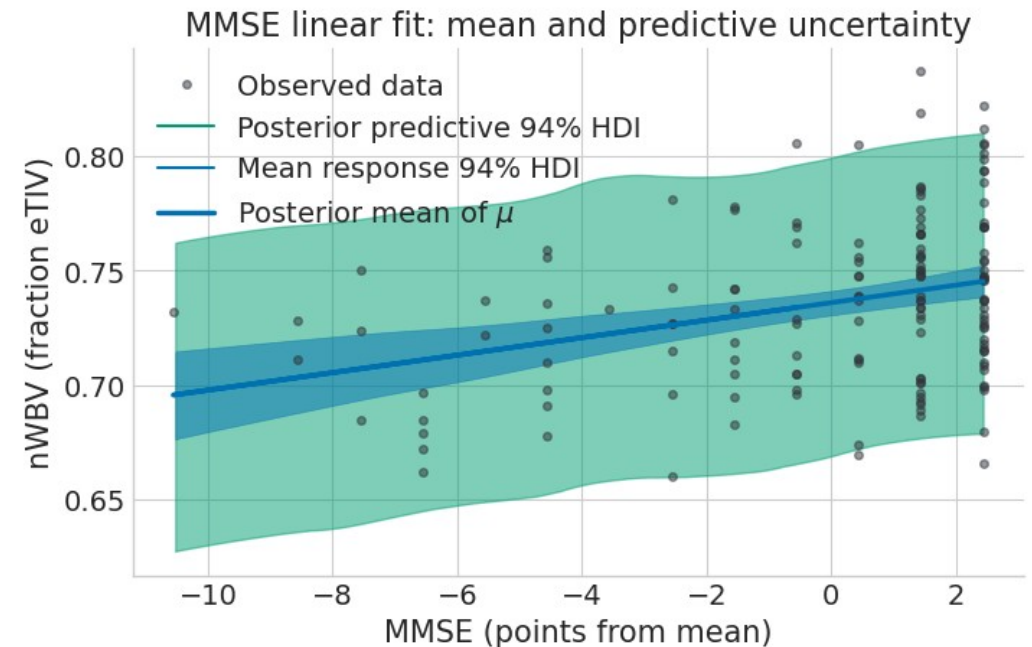
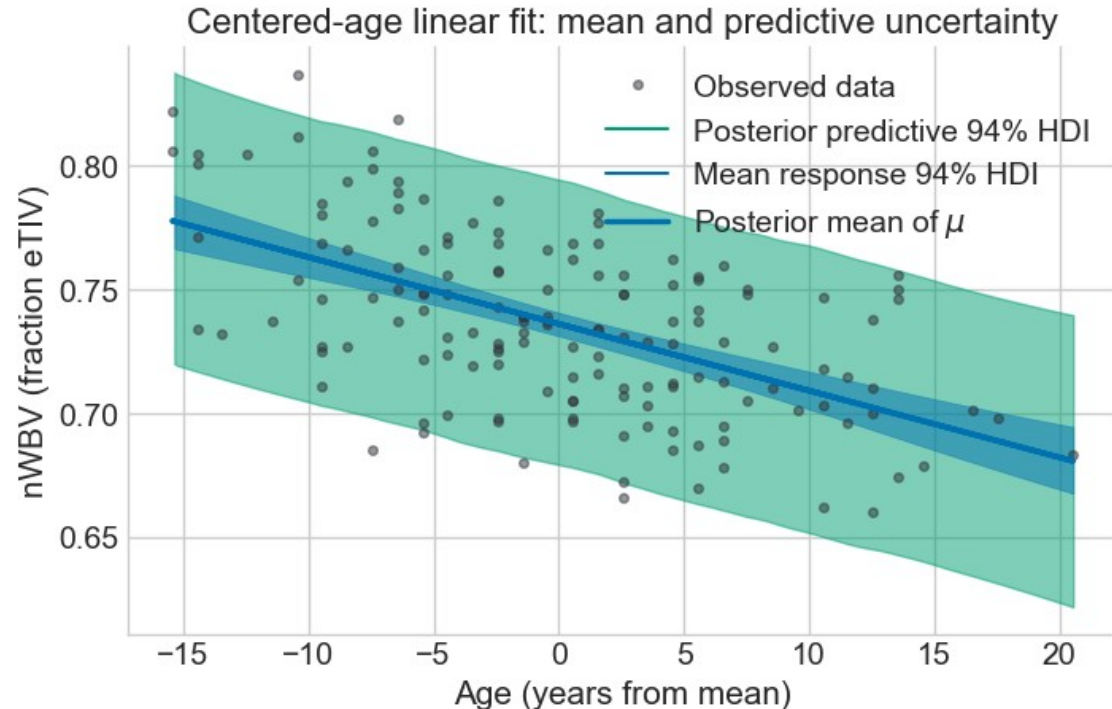
MMSE predicts brain volume

- Note that the MMSE is
 - Ordinal
 - Bounded
 - Non-Gaussian
- None of this violates the regression model
 - No assumptions about the independent variable

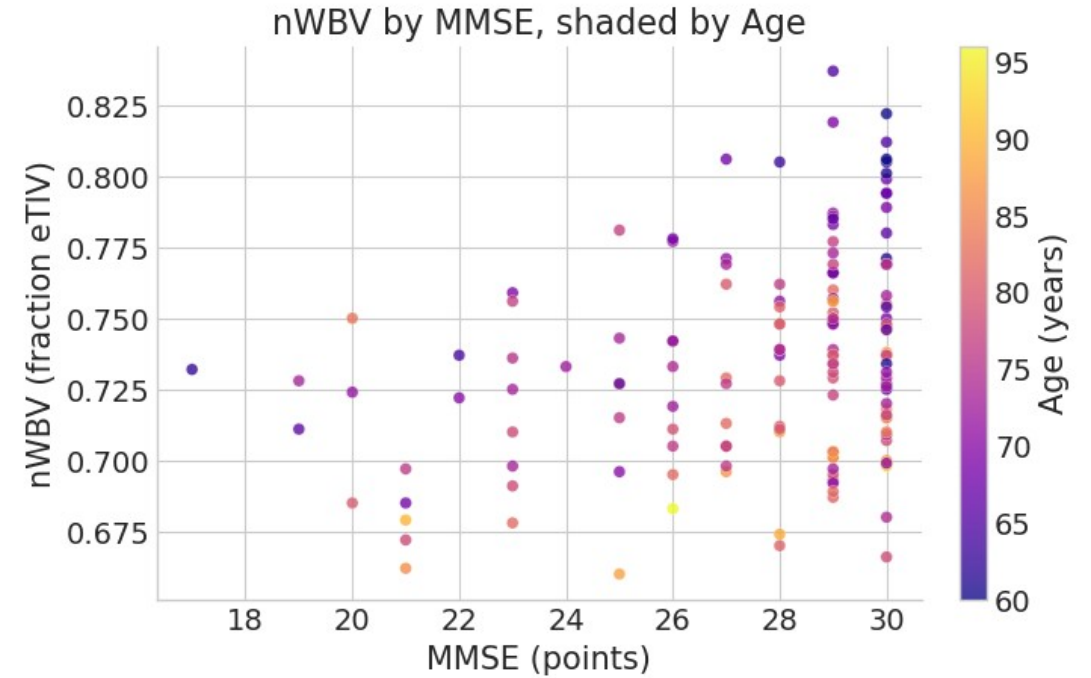
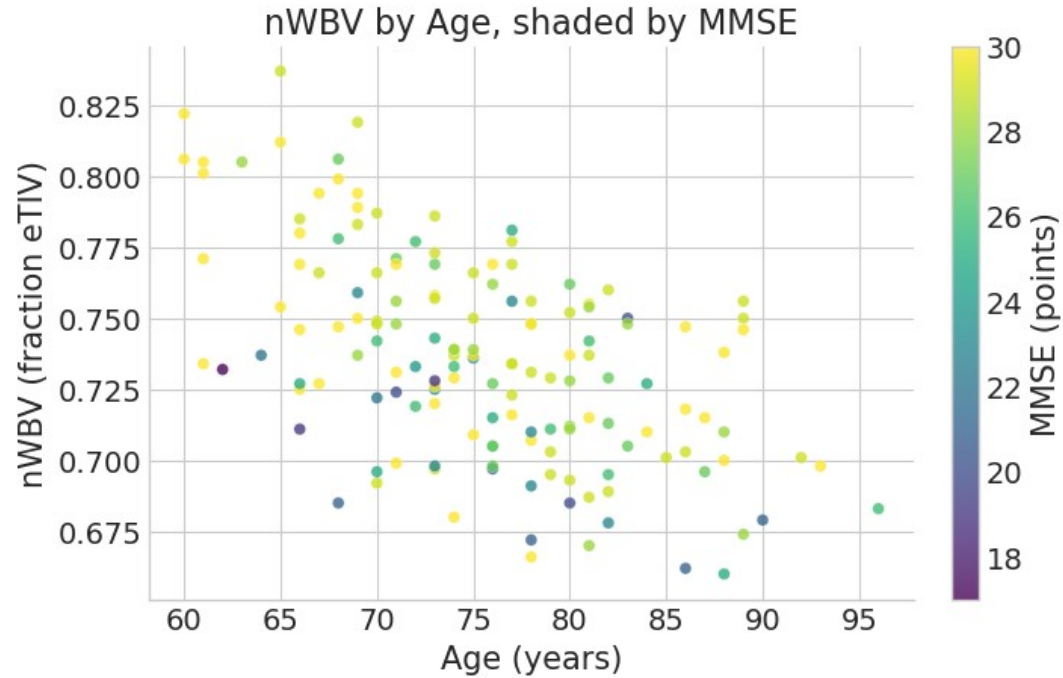


Can we ask about Age and MMSE together?

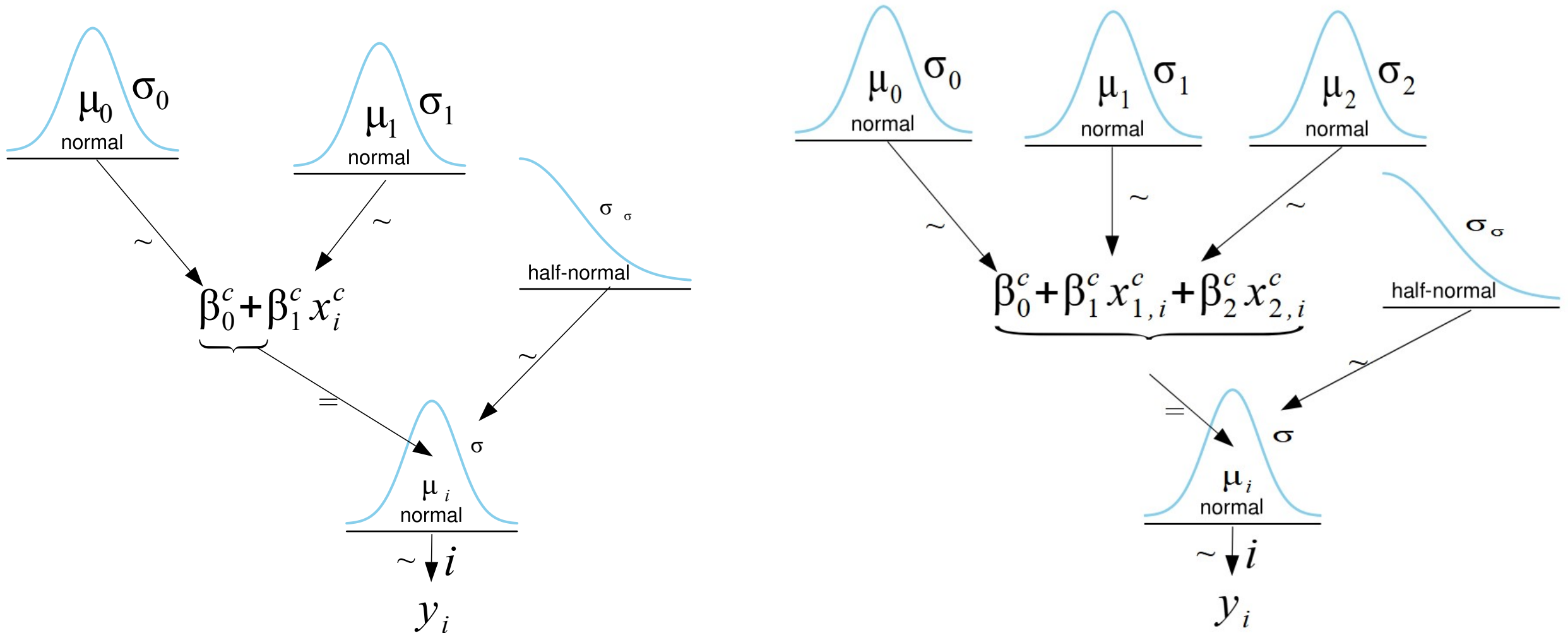
- Treat age and MMSE as two **independent** variables
- Treat nWBV as a **dependent** variable
- Combine into a single model



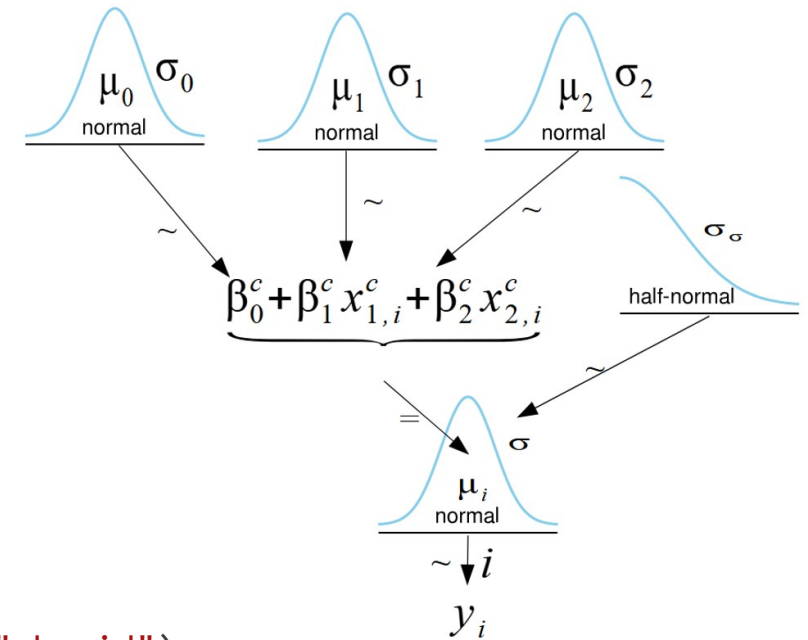
Three dimensional plots can help visualize



We generalize the bivariate model



And translate it to PyMC



```
with pm.Model(coords=coords_multiple) as nwbv_multiple_model:
    Age_c_data = pm.Data("Age_c_data", multiple_df["Age_c"].to_numpy(), dims="obs_id")
    MMSE_c_data = pm.Data("MMSE_c_data", multiple_df["MMSE_c"].to_numpy(), dims="obs_id")
    beta0 = pm.Normal("beta0", mu=mu_beta0, sigma=sigma_beta0)
    beta1 = pm.Normal("beta1", mu=mu_beta1, sigma=sigma_beta1)
    beta2 = pm.Normal("beta2", mu=mu_beta2, sigma=sigma_beta2)
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu = pm.Deterministic("mu", beta0 + beta1 * Age_c_data + beta2 * MMSE_c_data, dims="obs_id")
    nwbv_obs = pm.Normal("nwbv_obs", mu=mu, sigma=sigma, observed=multiple_df["nWBV"].to_numpy(), dims="obs_id")
```

We can generalize to matrix notation

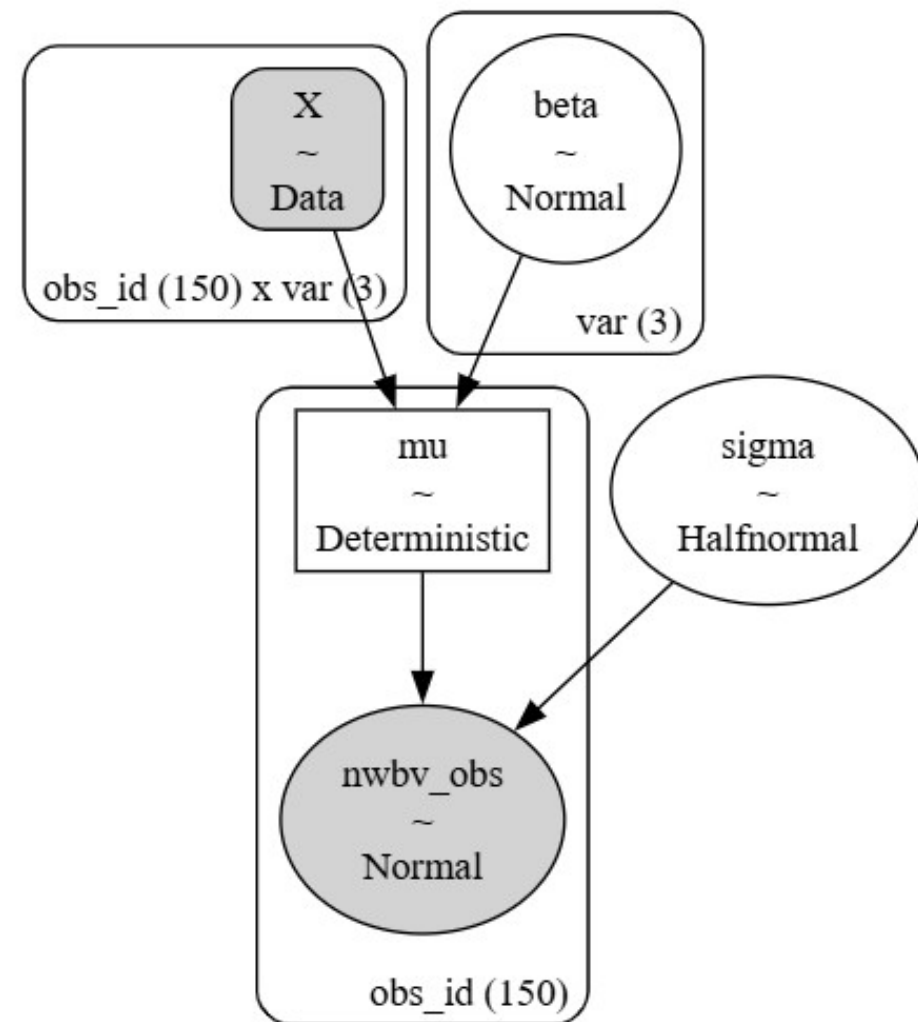
$$\mu_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} \rightarrow \boldsymbol{\mu} = \boldsymbol{\beta} \mathbf{X}$$

```
with pm.Model(coords=coords) as nwbv_multiple_model:
    Age_c_data = pm.Data("Age_c_data", multiple_df["Age_c"].to_numpy(), dims="obs_id")
    MMSE_c_data = pm.Data("MMSE_c_data", multiple_df["MMSE_c"].to_numpy(), dims="obs_id")
    beta0 = pm.Normal("beta0", mu=mu_beta0, sigma=sigma_beta0)
    beta1 = pm.Normal("beta1", mu=mu_beta1, sigma=sigma_beta1)
    beta2 = pm.Normal("beta2", mu=mu_beta2, sigma=sigma_beta2)
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu = pm.Deterministic("mu", beta0 + beta1 * Age_c_data + beta2 * MMSE_c_data, dims="obs_id")
    nwbv_obs = pm.Normal("nwbv_obs", mu=mu, sigma=sigma, observed=multiple_df["nWBV"].to_numpy(), dims="obs_id")
```

```
with pm.Model(coords=coords) as model_matrix:
    X = pm.Data("X", X_matrix, dims=("obs_id", "var"))
    beta = pm.Normal("beta", mu=mu_beta, sigma=sigma_beta, dims="var")
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu = pm.Deterministic("mu", pm.math.dot(X, beta), dims="obs_id")
    nwbv_obs = pm.Normal(
        "nwbv_obs",
        mu=mu,
        sigma=sigma,
        observed=multiple_df["nWBV"].to_numpy(),
        dims="obs_id",
    )
```

We can generalize to matrix notation

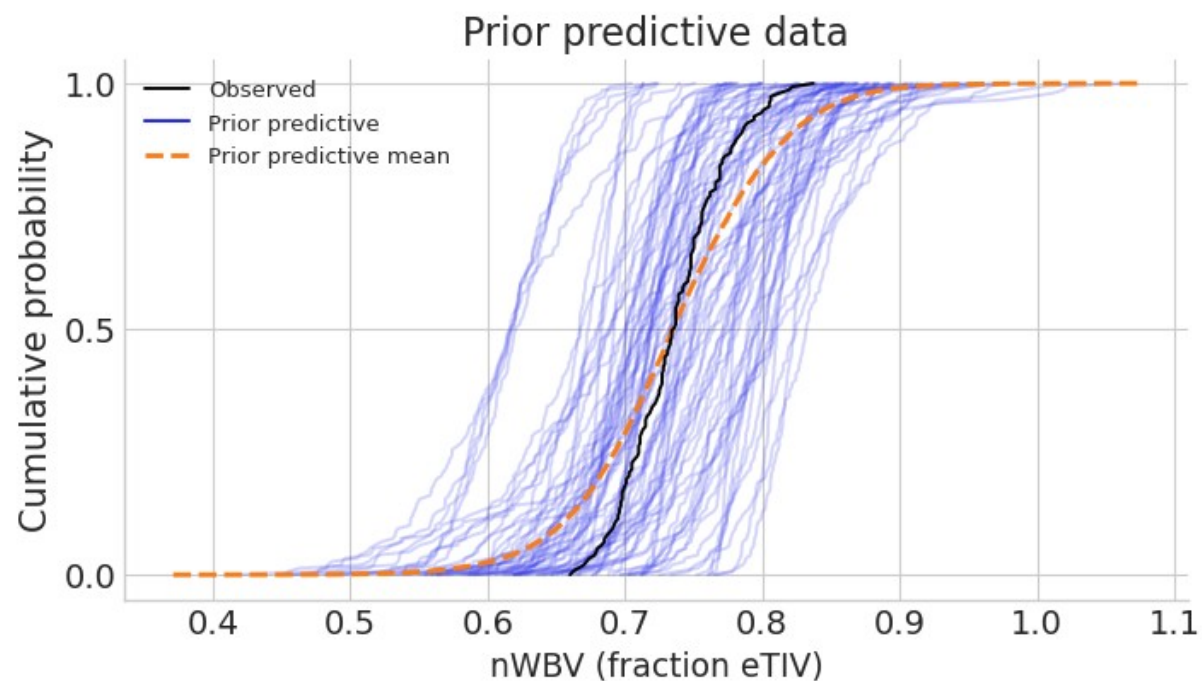
```
with pm.Model(coords=coords) as model_matrix:  
    X = pm.Data("X", X_matrix, dims=("obs_id", "var"))  
    beta = pm.Normal("beta", mu=mu_beta, sigma=sigma_beta, dims="var")  
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)  
    mu = pm.Deterministic("mu", pm.math.dot(X, beta), dims="obs_id")  
    nwbv_obs = pm.Normal(  
        "nwbv_obs",  
        mu=mu,  
        sigma=sigma,  
        observed=multiple_df["nWBV"].to_numpy(),  
        dims="obs_id",  
    )
```



The data matrix holds all the independent variables

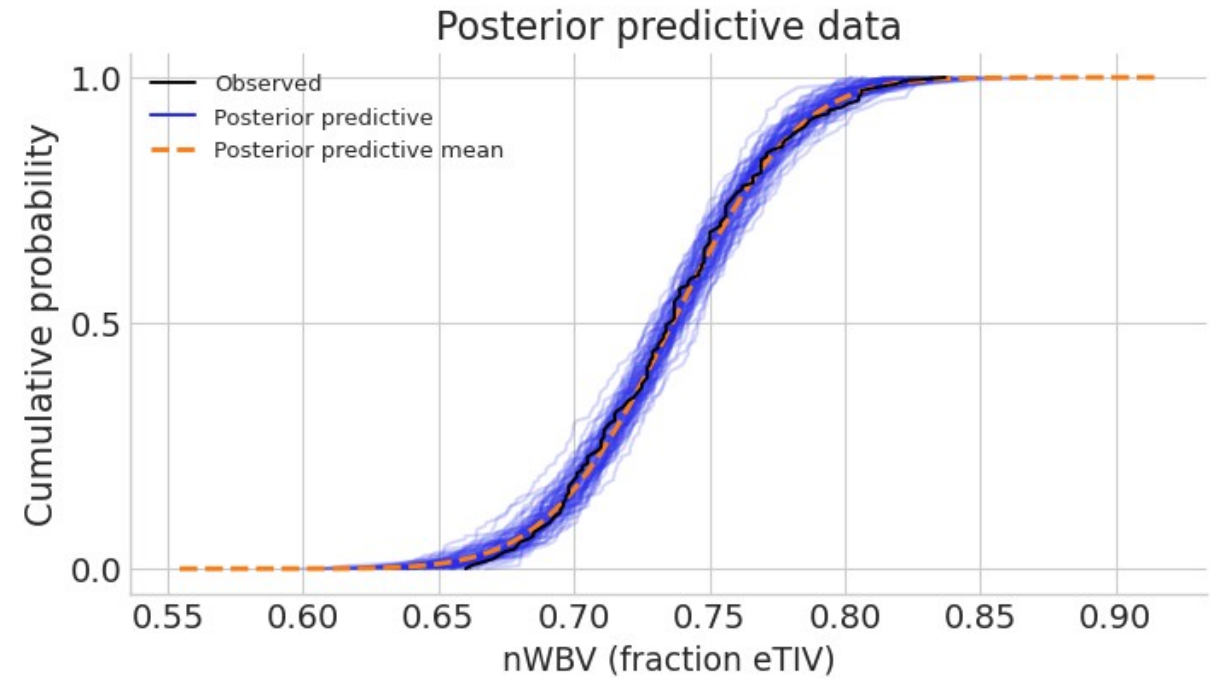
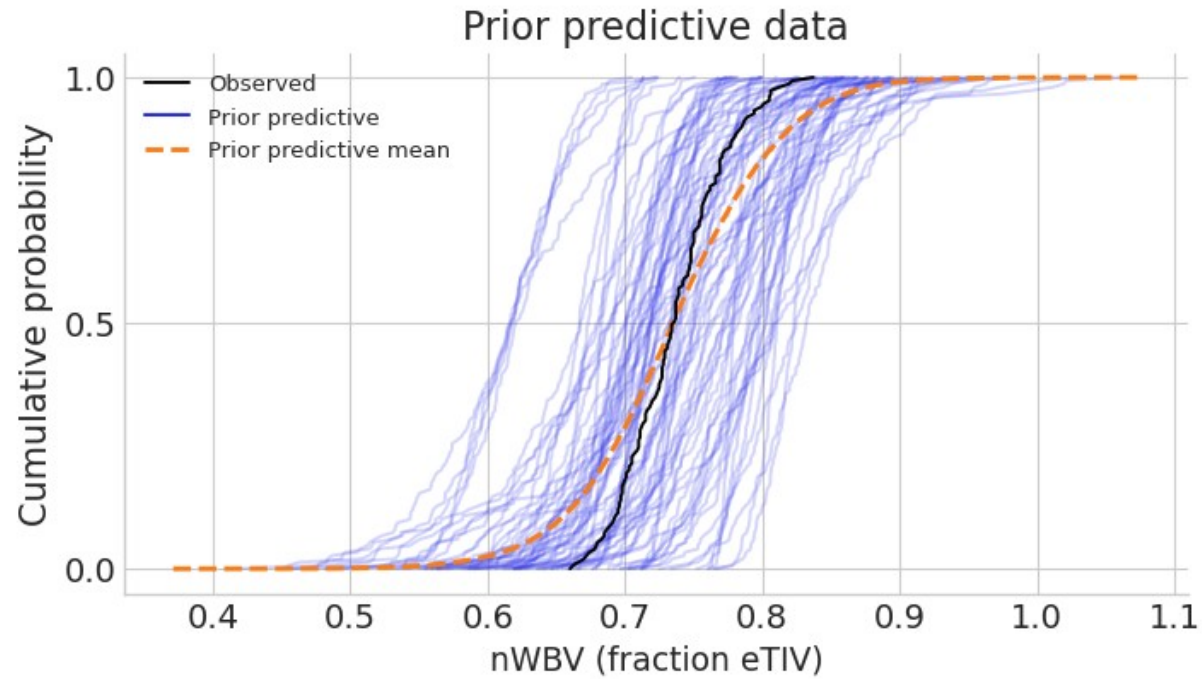
$$\mathbf{X} = \begin{pmatrix} 1 & x_{1,1} & \cdots & x_{k,1} \\ 1 & x_{2,1} & \cdots & x_{k,2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{2,N} & \cdots & x_{k,N} \end{pmatrix} \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} \quad \boldsymbol{\mu} = \begin{pmatrix} \mu_0 \\ \mu_1 \\ \vdots \\ \mu_N \end{pmatrix} = \mathbf{X}\boldsymbol{\beta}$$

Always do the workflow



	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
beta[intercept]	0.736	0.002	0.732	0.740	0.0	0.0	4288.0	2777.0	1.0
beta[Age_c]	-0.003	0.000	-0.003	-0.002	0.0	0.0	5362.0	2592.0	1.0
beta[MMSE_c]	0.004	0.001	0.002	0.005	0.0	0.0	5582.0	3085.0	1.0
sigma	0.028	0.002	0.025	0.031	0.0	0.0	4650.0	2816.0	1.0

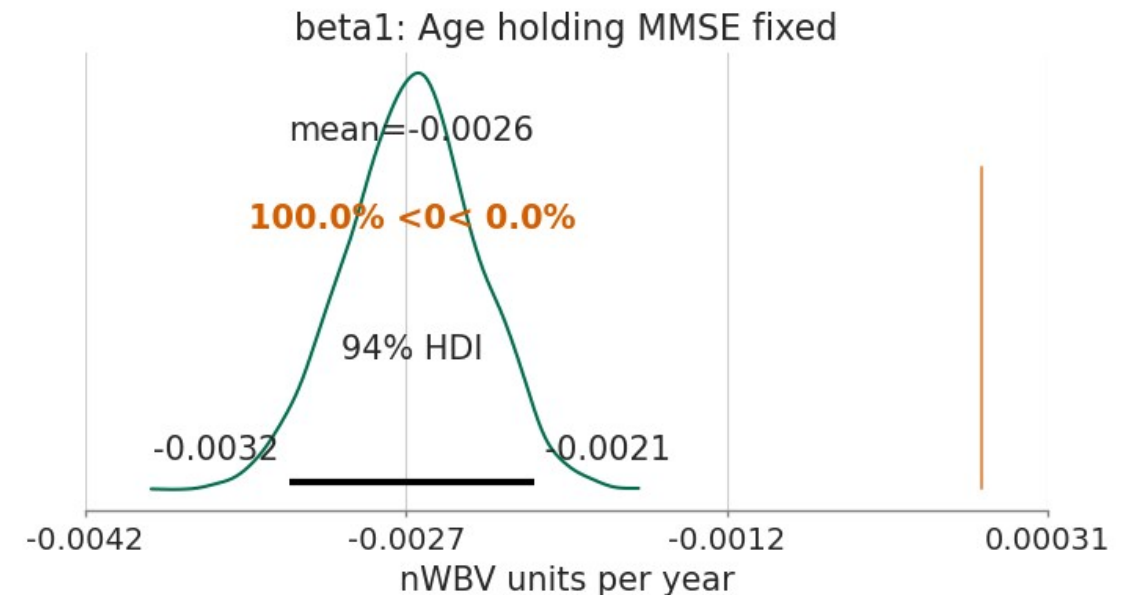
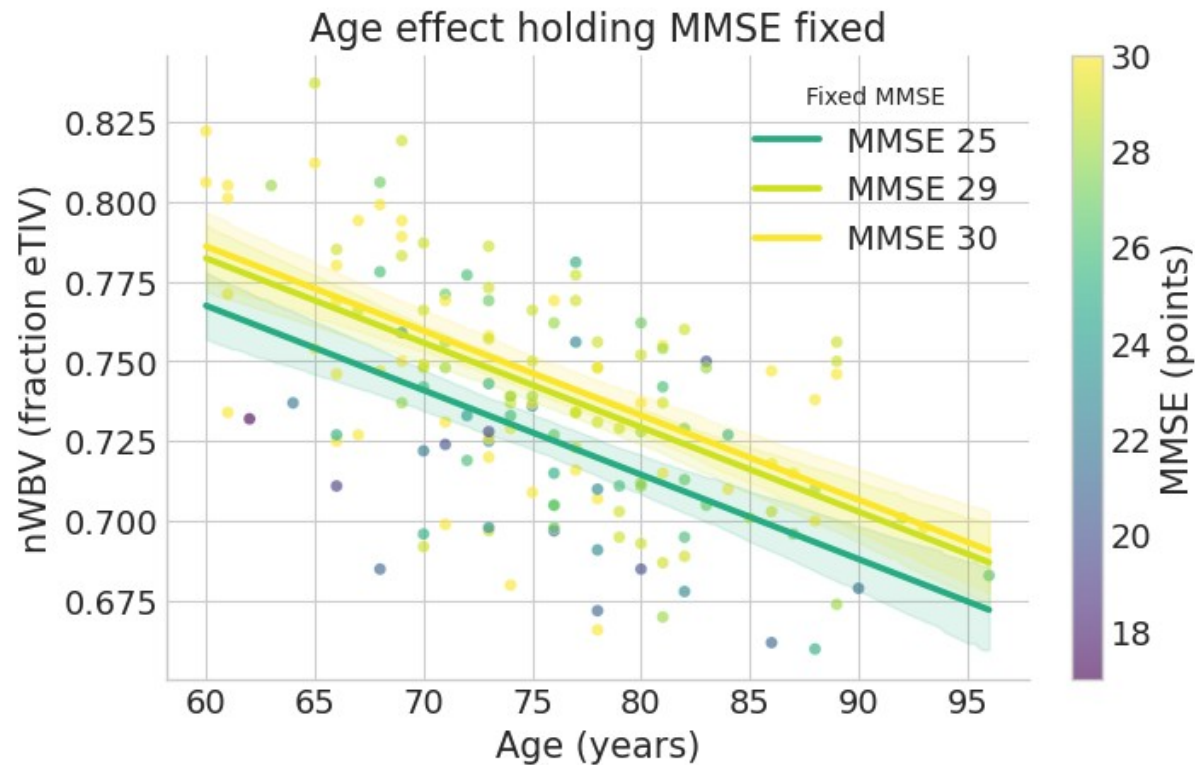
Check goodness of fit



8.6 Multivariate regression posterior

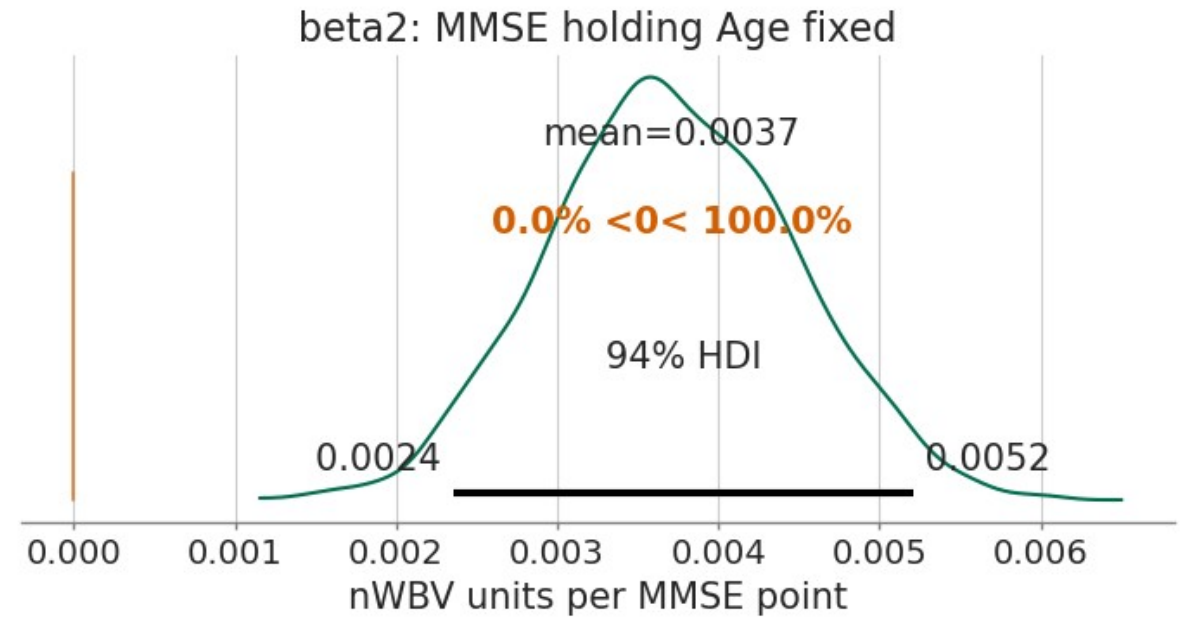
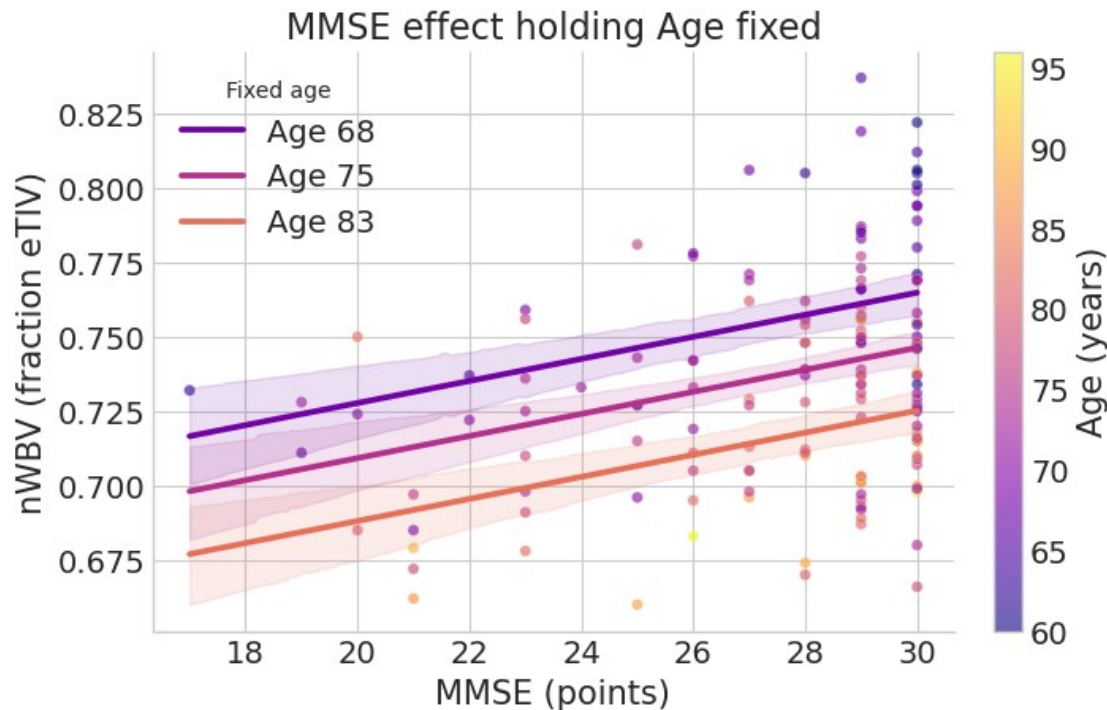
β_{Age} gives the effect of Age when MMSE is held fixed

- nWBV is reduced around 0.25% / year
 - HDI is 0.2%-0.3%
 - For different values of MMSE



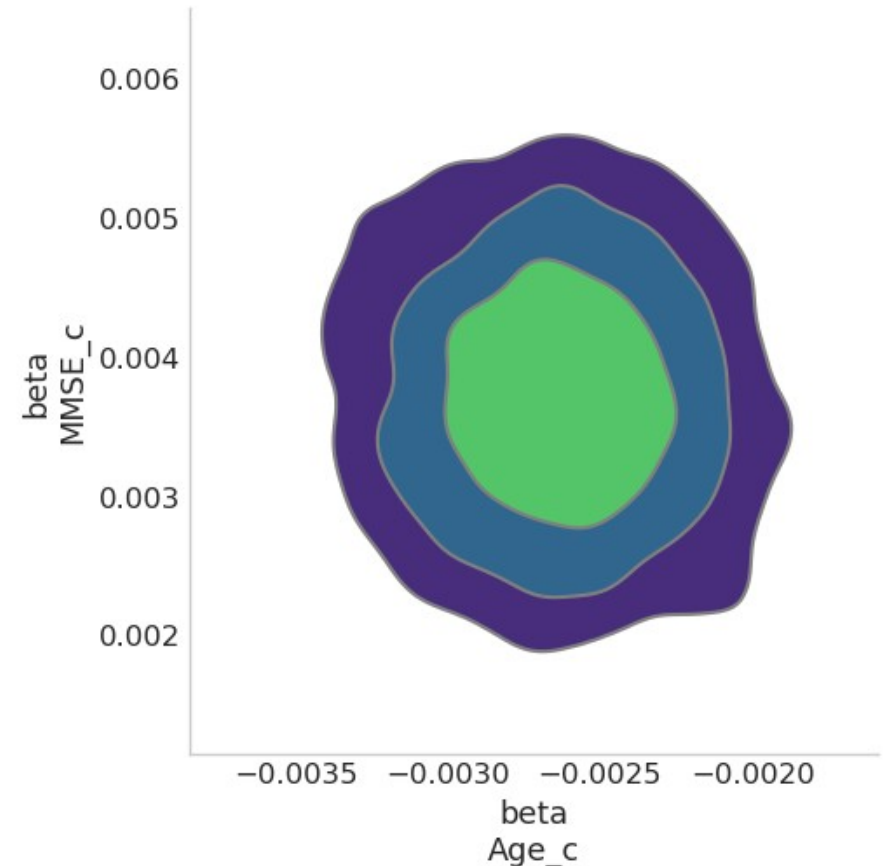
β_{MMSE} gives the effect of MMSE when Age is held fixed

- nWBV goes up by 0.4% for every point of MMSE
 - HDI is 2.4%-5.2%
 - For a given age



The two slope parameters are not correlated

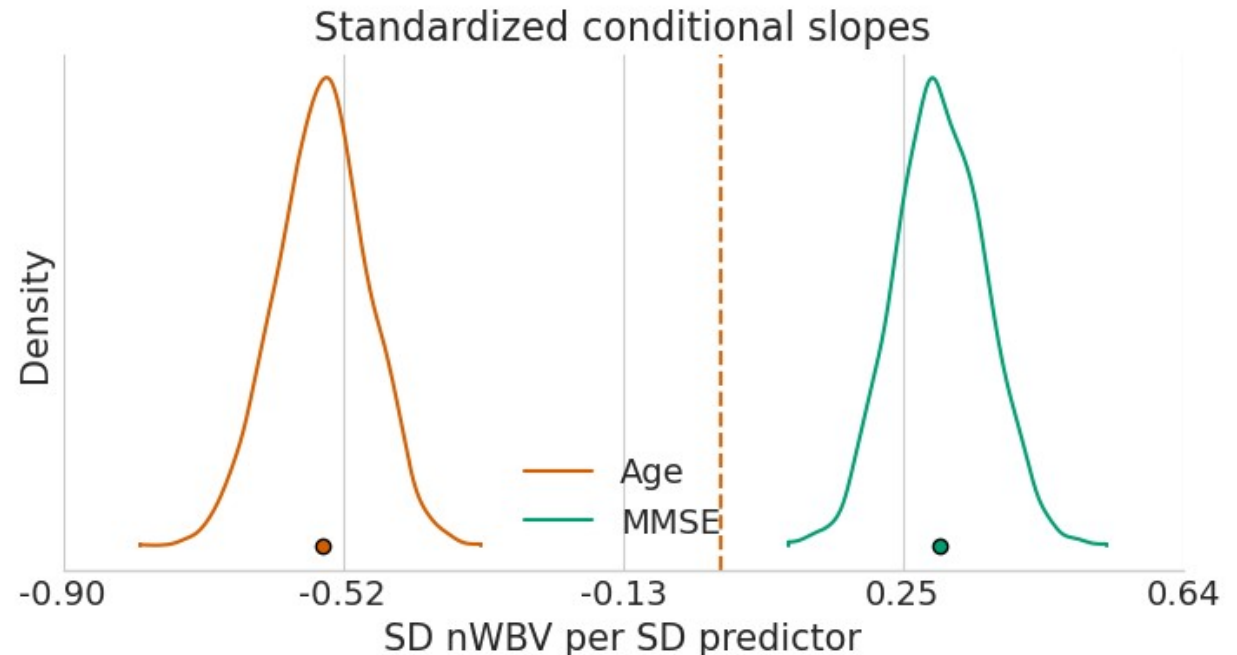
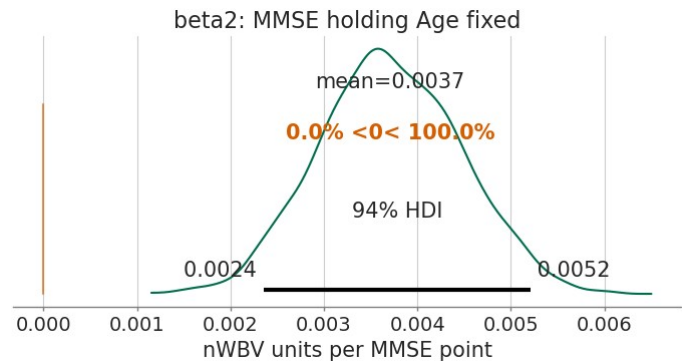
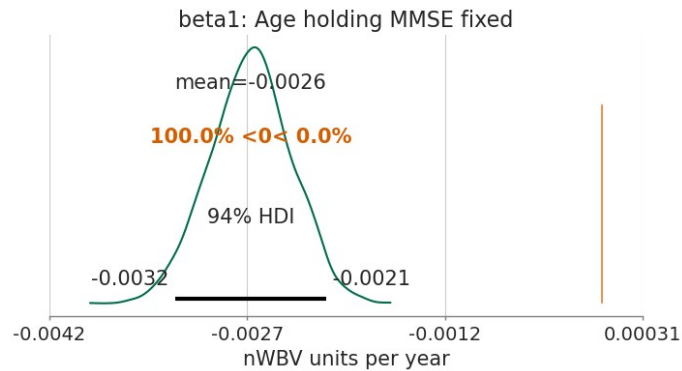
- Correlated parameters can be a problem for models
 - Like with uncentered data



Compare effect sizes using standardized slopes

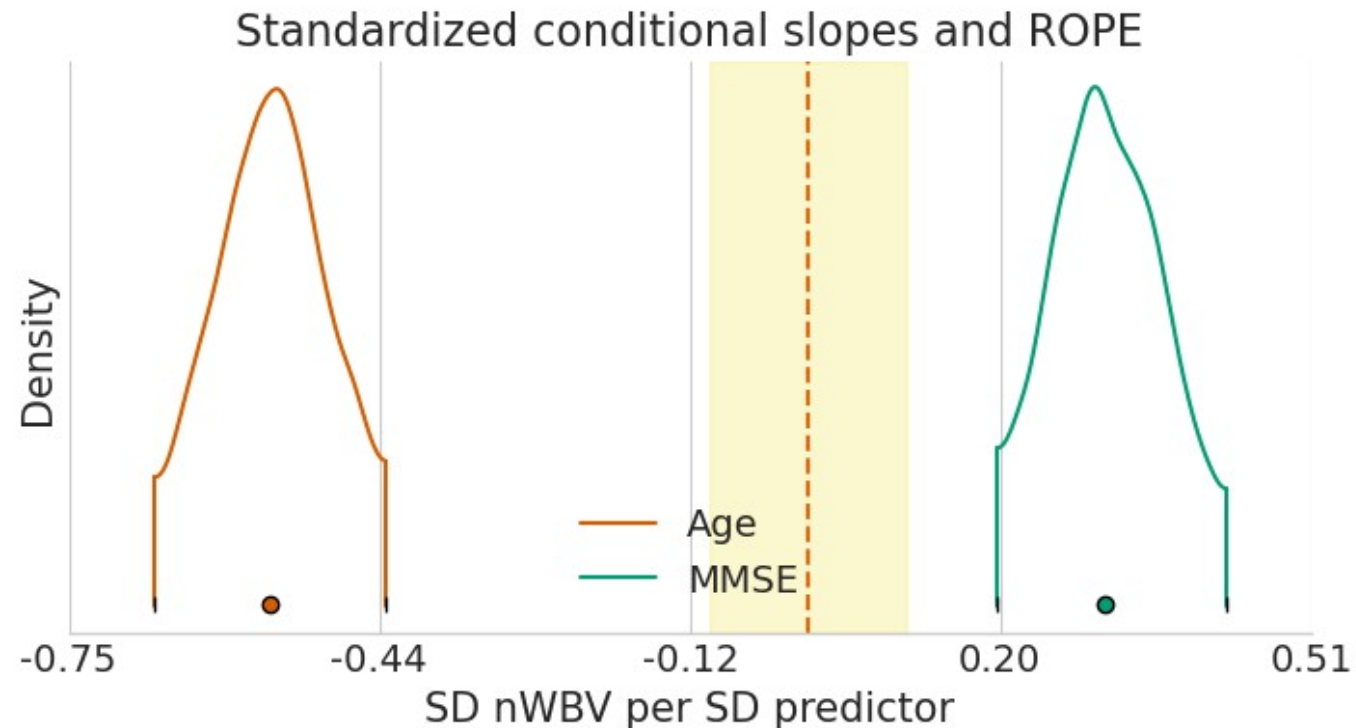
- Compare slopes despite difference in units

$$\beta_k^z = \frac{s_{x_k}}{s_y} \beta_k$$



Compare HDI and ROPE on standardized effects

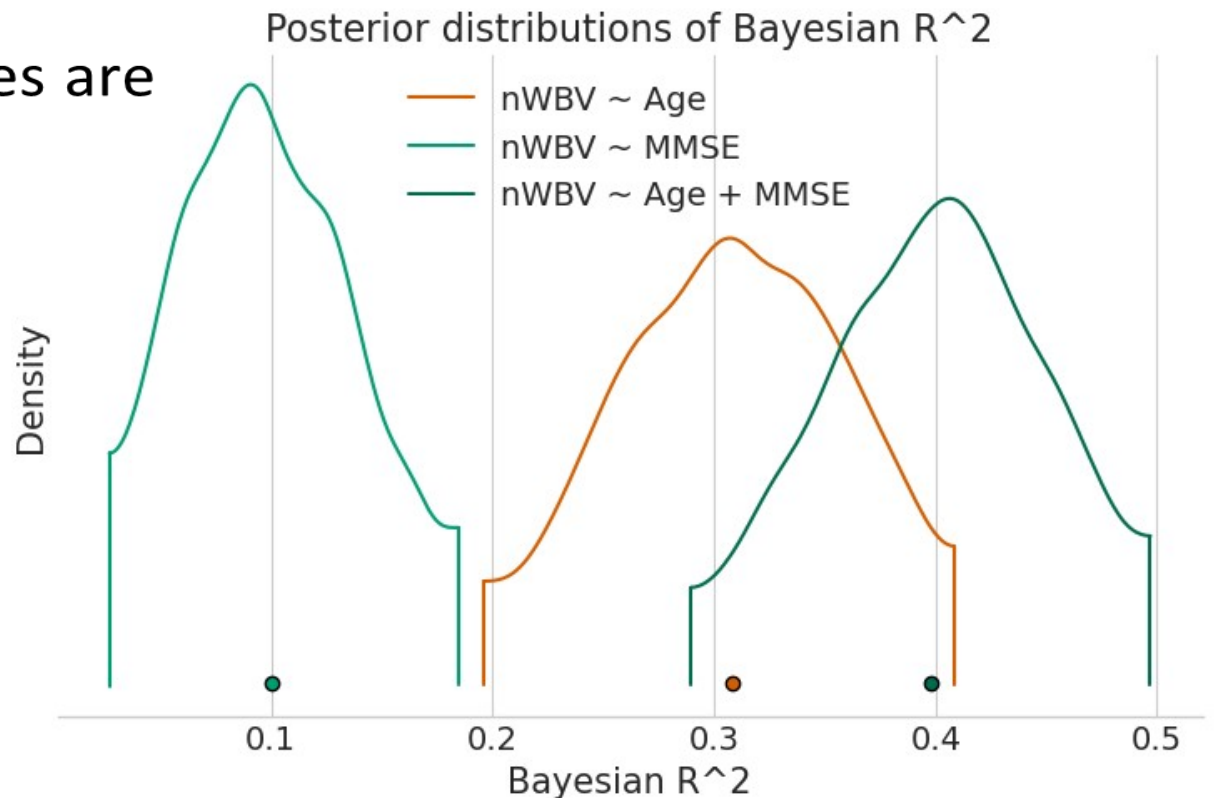
- Plots show only the HDI without the tails
- No overlap with ROPE



Variance explained is used as an effect size

- R^2 divides the variance into two parts:
 - Variance captured by the model: $\text{var}(\mu_i)$
 - Residual variance in the data: σ^2
 - In general: $\text{var}(y) = \text{var}(\mu_i) + \sigma^2$
 - Although not exactly
- R^2 always increases when new variables are added to the model

$$R^2 = \frac{\text{var}(\mu_i)}{\text{var}(\mu_i) + \sigma^2}$$



8.7 Frequentist methods

Frequentist multivariate model, estimators for regression coefficients, effect sizes, hypothesis tests

The frequentist multivariate regression model

- Estimators use Roman letters instead of Greek letters
- The estimator can be derived by minimizing the sum squared error

$$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$$

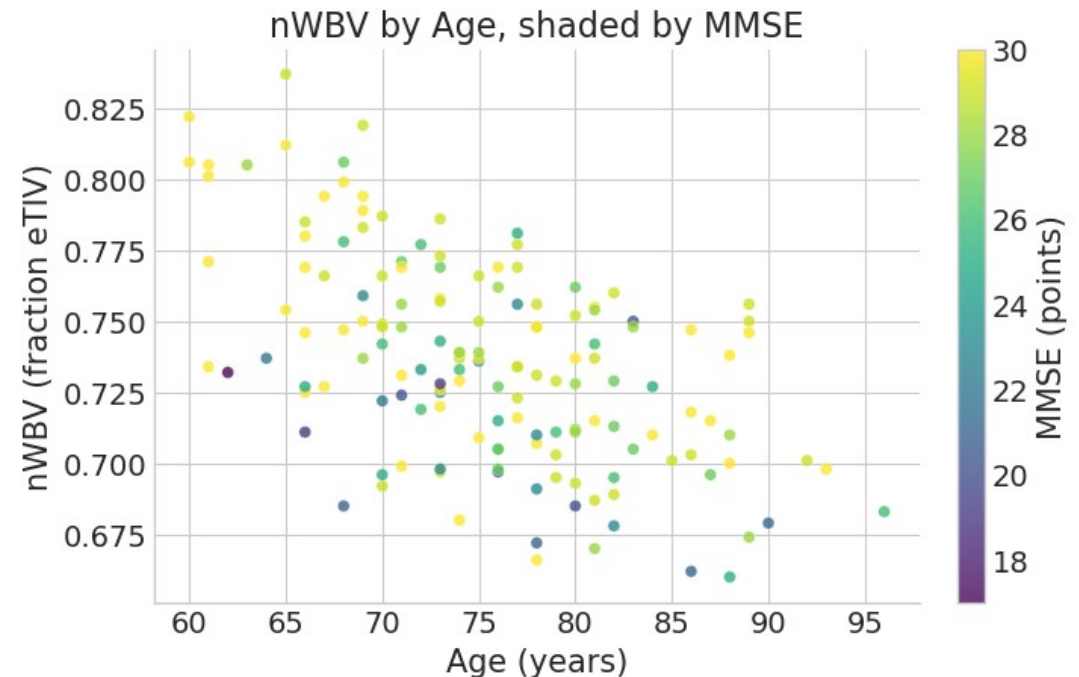
$$\mathbf{y} \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$$

Estimators:

$$\hat{\mathbf{y}} = \mathbf{bX}$$

Minimize:

$$(\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}}) = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$



The estimators have a simple formula

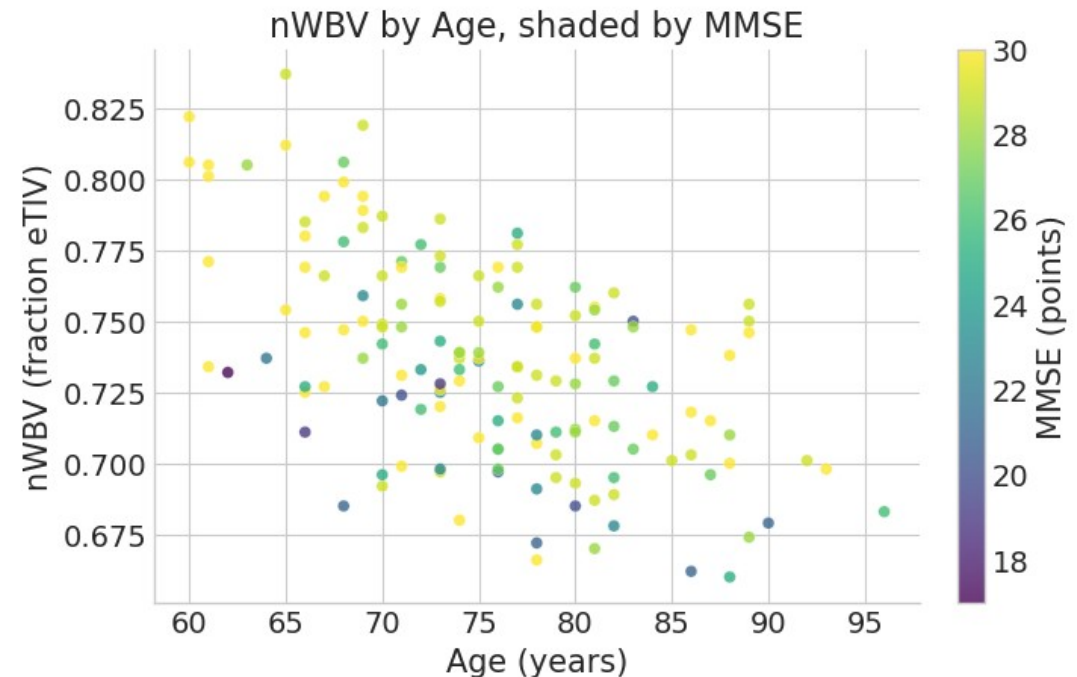
- $\mathbf{X}$ times $\mathbf{y}$
- Over $\mathbf{X}$ times $\mathbf{X}$

$$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta} \leftrightarrow \hat{\mathbf{y}} = \mathbf{X}\mathbf{b}$$

$$\mathbf{y} \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$$

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$s^2 = (\mathbf{y} - \mathbf{X}\mathbf{b})^T (\mathbf{y} - \mathbf{X}\mathbf{b})$$



The sampling distribution of the estimators is normal

- They are unbiased
- With known variance
- That can be estimated
- So each estimator b_k is normally distributed with known variance that can be estimated

$$\boldsymbol{\mu} = \boldsymbol{\beta} \mathbf{X} \leftrightarrow \hat{\mathbf{y}} = \mathbf{b} \mathbf{X}$$

$$\mathbf{y} \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$$

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X} \mathbf{y}$$

$$s^2 = (\mathbf{y} - \mathbf{X} \mathbf{b})^T (\mathbf{y} - \mathbf{X} \mathbf{b})$$

$$\mathbf{b} \sim N(\boldsymbol{\beta}, \boldsymbol{\Sigma}_b)$$

$$\boldsymbol{\Sigma}_b = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

$$\mathbf{s}_b^2 = s^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

From the sampling distribution we can do statistics

- Each b_k is normally distributed
 - So we can get a confidence interval
- b_i/s_{b_i} is distributed with a Student's t distribution
 - Under the null hypothesis
 - With $N - (k + 1)$ degrees of freedom
 - If there are k independent variables
 - So we can do a hypothesis test

term	estimate	SE	standardized slope	t	p	95% CI lower	95% CI upper
b0	0.7361	0.0023	—	318.985	<0.0001	0.7315	0.7406
b1: Age_c	-0.0027	0.0003	-0.551	-8.675	<0.0001	-0.0033	-0.0021
b2: MMSE_c	0.0038	0.0008	0.310	4.873	<0.0001	0.0023	0.0053

The standard error (SE) is the standard deviation of the sampling distribution.

Compare frequentist to Bayesian intervals

term	estimate	SE	standardized slope	t	p	95% CI lower	95% CI upper
b0	0.7361	0.0023	—	318.985	<0.0001	0.7315	0.7406
b1: Age_c	-0.0027	0.0003	-0.551	-8.675	<0.0001	-0.0033	-0.0021
b2: MMSE_c	0.0038	0.0008	0.310	4.873	<0.0001	0.0023	0.0053

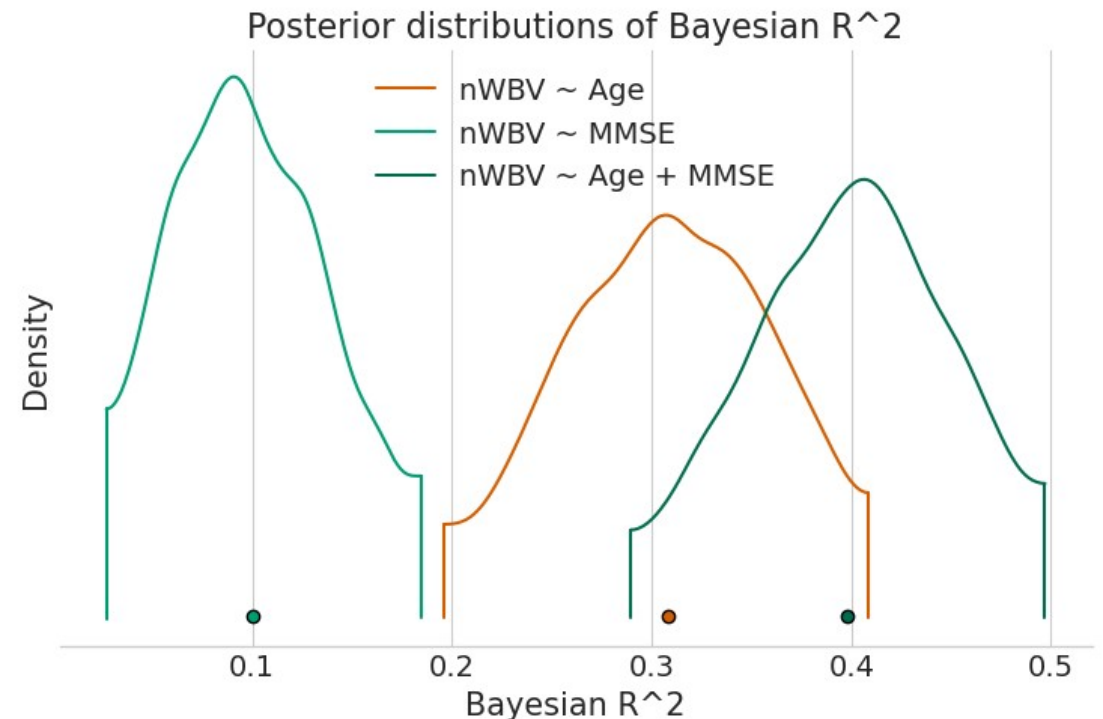
Predictor	Raw effect mean	Standardized effect mean	Std. 94% HDI lower	Std. 94% HDI upper	Posterior in ROPE	HDI overlaps ROPE
Age	-0.0026	-0.547	-0.664	-0.429	0.000	no
MMSE	0.0037	0.302	0.192	0.424	0.001	no

Frequentist R^2 to as effect size for regression

- Variance explained over total variance
 - But use only the estimators
 - Without a distribution

$$R^2 = \frac{\text{var}(\hat{y}_i)}{\text{var}(y_i)} = 0.407$$

Model	R^2
nWBV ~ Age	0.311
nWBV ~ MMSE	0.104
nWBV ~ Age + MMSE	0.407



Define some useful statistics

- The total variability in the model
 - The sum squared distance between the model and the mean
- The total variability not explained by the model
 - The squared distance between the data and the model

$$SS_{\text{reg}} = (\hat{\mathbf{y}} - \bar{y})^T (\hat{\mathbf{y}} - \bar{y}) = \sum_{i=1}^N (\hat{y}_i - \bar{y})^2$$

$$SS_{\text{res}} = (\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}}) = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Count the degrees of freedom

- There is a real logic to counting degrees of freedom
- We will not learn it
- k independent variables

$$SS_{\text{reg}} = (\hat{\mathbf{y}} - \bar{y})^T (\hat{\mathbf{y}} - \bar{y}) = \sum_{i=1}^N (\hat{y}_i - \bar{y})^2$$

$$df_{\text{reg}} = k$$

$$SS_{\text{res}} = (\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}}) = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$df_{\text{res}} = N - (k + 1)$$

The F statistic can be used for hypothesis tests

- The F statistic is a ratio of two measures of variance
 - Variance from the model in the numerator
 - Variance for the residuals in the denominator
- Under the null model, these two variances should be approximately equal
 - So the ratio is around 1
 - With known distribution called the F distribution
 - Paramaterized by the two degrees of freedom

$$F = \frac{MS_{reg}}{MS_{res}} = \frac{\frac{SS_{reg}}{df_{reg}}}{\frac{SS_{res}}{df_{res}}} \sim F(df_{reg}, df_{res})$$

Model-level null	F statistic	p-value
H0: beta1 = beta2 = 0	50.482	<0.0001

8.8 Next time

Combining regression with categorical variables